

Average, Total, and Concentration Hierarchy Indices

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Abstract

This paper studies numerical measures of hierarchy on directed acyclic graphs. It distinguishes hierarchy per person, aggregate hierarchy, and hierarchy concentration. These interpretations correspond to average, total, and pairwise population normalizations and imply distinct replication rules. The analysis considers indices that aggregate local scores based on individual depth and span. No depth–span average index satisfies both the unrestricted upward extension and subordination removal axioms. Under coordinatewise monotonicity and uniform strictness in at least one coordinate, however, average indices satisfy conditional extension axioms and subordination removal; an additional boundary condition ensures unrestricted downward extension. Any depth–span concentration index that is positive on some hierarchy violates all unrestricted and conditional extension axioms, without restrictions on its local evaluation function. By contrast, total indices with nonnegative, coordinatewise monotone local evaluations that are uniformly strict in at least one coordinate satisfy extensive replication, subordination removal, and all extension axioms. Population normalization therefore determines both replication behavior and compatibility with structural monotonicity.

Keywords: hierarchy measurement, axiomatic characterization, directed acyclic graph, depth and span, population normalization, organizations.

JEL classifications: D23, L22.

1 Introduction

Hierarchical structures are a defining feature of economic and social organizations. An individual’s position within these structures shapes their access to information, influence, and opportunity (Uzzi, 1997; Brass, 1984; Podolny, 2001; Zuckerman, 1999). Hierarchies can enhance coordination and efficiency in uncertain environments (Simon, 1962, 1973; Holland, 1998; Lawrence and Lorsch, 1967; Thompson, 1967; Lincoln and Kalleberg, 1985), and informal reporting networks can strengthen organizational resilience (Krackhardt and Stern, 1988; Sarkar et al., 2010; Culen, 2017). At the same time, hierarchies can entrench inequality and restrict information flow (Anderson, 1972; Fix, 2017, 2018, 2019, 2021; Wright, 2024). These considerations highlight the need for well-founded numerical measures of hierarchy.

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Prior axiomatic work establishes ordinal frameworks for ranking hierarchies (Carbonell-Nicolau, 2025a,b), but does not distinguish how measured hierarchy should scale with organizational size. Consider the disjoint union of several identical copies of a hierarchy. Replication preserves each copy’s internal structure while increasing both the total population and the number of disconnected components. Its measurement therefore depends on whether hierarchy is interpreted per person, in aggregate, or relative to population size. These interpretations yield three mutually incompatible scaling principles:

1. **Hierarchy per person** evaluates the average structural position within an organization. It requires replication invariance and supports comparisons of per-capita administrative structure across organizations of different sizes.
2. **Aggregate hierarchy** evaluates the total volume of hierarchical structure. It requires the index to increase under replication and supports comparisons of the aggregate scale of hierarchy. The total indices studied below scale linearly under replication.
3. **Concentration of hierarchy** evaluates hierarchical structure relative to the number of potential pairs of individuals. Because the number of potential pairs grows quadratically with population, replication lowers any positive index of this form. This interpretation measures hierarchy relative to population size rather than per person or in aggregate.

To translate these scaling principles into measurement, we study indices that aggregate local scores based on individual depth (number of supervisors) and span (number of subordinates) on directed acyclic graphs. The framework separates the choice of local evaluation from the choice of population normalization. The three interpretations are implemented through different population normalizations: dividing the sum of local scores by the population size (hierarchy per person), applying no population normalization (aggregate hierarchy), or dividing by the number of potential unordered pairs of individuals (concentration of hierarchy).

Beyond replication, a robust measurement framework must account for vertical growth. We introduce unrestricted extension axioms requiring an index to strictly increase whenever management layers are appended above or below an existing structure. We also consider conditional variants that permit partial attachments when the appended hierarchy has an index value at least as high as the base hierarchy.

Our analysis reveals a tension between population normalization and vertical extension. We prove that no depth–span average index satisfies both the unrestricted upward extension and subordination removal axioms. The conflict arises from the interaction of population averaging with the depth–span restriction: large flat upper appendages can dilute the population average. Under coordinatewise monotonicity and uniform strictness in at least one coordinate, however, average indices satisfy both conditional extension axioms and subordination removal; an additional boundary condition ensures unrestricted downward extension.

For concentration indices, the incompatibility is stronger: any depth–span concentration index that is positive on some hierarchy violates every unrestricted and conditional extension axiom. This impossibility requires no monotonicity, sign, or growth restrictions on the local evaluation function; it is driven by the dilution inherent in normalizing aggregate hierarchy by the number of potential pairs.

By contrast, depth–span total indices avoid these dilution effects. If the local evaluation has a zero baseline, is weakly increasing in both coordinates, and is uniformly strictly increasing in at least one coordinate, the resulting index satisfies extensive replication, subordination removal, and all extension axioms.

Applying the framework to established indices shows that they capture different measurement objectives and often fail the present axioms. The global reaching centrality index (Mones et al., 2012) satisfies anonymity, but its reliance on a global reference point generates non-monotonic behavior that violates core extension and subordination removal axioms. On directed acyclic graphs, Krackhardt's (1994) graph hierarchy index is constant, while his other dimensions violate several of the present axioms. Other measures (Trusina et al., 2004; Corominas-Murtra et al., 2013; Luo and Magee, 2011) are calibrated primarily to detect cycles, collapsing to uninformative values on acyclic structures. Finally, algorithmic approaches (Czégel and Palla, 2015) depend on exogenous decay parameters that can violate structural monotonicity.

The remainder of the paper proceeds as follows. Section 2 formalizes hierarchies as directed acyclic graphs. Section 3 develops the axiomatic framework, including the replication principles and vertical extension axioms. Section 4 presents the core theoretical results, characterizing the average, concentration, and total index families and establishing their compatibility or incompatibility with structural extension. Section 5 applies the framework to indices from the literature. Section 6 concludes.

2 Hierarchies

A *hierarchy* is a finite, nonempty directed acyclic graph

$$h = (V(h), E(h)).$$

The nodes in $V(h)$ represent individuals, and an edge $(j, i) \in E(h)$ means that j directly supervises i . We write $j \rightarrow i$ whenever $(j, i) \in E(h)$. Acyclicity rules out directed paths from a node to itself.

For each $v \in V(h)$, define

$$\text{Anc}_h(v) = \{u \in V(h) : u \rightsquigarrow v\}$$

and

$$\text{Desc}_h(v) = \{w \in V(h) : v \rightsquigarrow w\},$$

where $u \rightsquigarrow v$ means that there is a directed path of positive length from u to v . Thus, $\text{Anc}_h(v)$ is the set of direct and indirect supervisors of v , while $\text{Desc}_h(v)$ is the set of its direct and indirect subordinates. Because paths have positive length and h is acyclic,

$$v \notin \text{Anc}_h(v) \quad \text{and} \quad v \notin \text{Desc}_h(v).$$

The *depth* and *span* of v are, respectively,

$$s_h(v) = |\text{Anc}_h(v)| \quad \text{and} \quad \sigma_h(v) = |\text{Desc}_h(v)|.$$

Depth counts the supervisors of a node, while span counts its subordinates.

If $j \rightarrow i$, then every ancestor of j is also an ancestor of i . Hence, $\text{Anc}_h(j) \subseteq \text{Anc}_h(i)$. Moreover,

$$j \in \text{Anc}_h(i) \setminus \text{Anc}_h(j).$$

Therefore, $s_h(j) < s_h(i)$. More generally, if $j \rightsquigarrow i$, then

$$\text{Anc}_h(j) \subsetneq \text{Anc}_h(i),$$

and consequently $s_h(j) < s_h(i)$. Thus, every direct or indirect supervisor of a node has strictly smaller depth than that node.

A *root* has in-degree zero and hence no supervisors; a *leaf* has out-degree zero and hence no subordinates. The *size* of h is $|V(h)|$.

Let $\mathcal{H}$ denote the class of all finite, nonempty hierarchies. For each integer $n \geq 1$, let

$$\mathcal{H}_n = \{h \in \mathcal{H} : |V(h)| = n\}.$$

Then

$$\mathcal{H} = \bigcup_{n \geq 1} \mathcal{H}_n.$$

Hierarchies $h_1, \dots, h_k$ are *node-disjoint* if their node sets are pairwise disjoint. For node-disjoint hierarchies $h_1, \dots, h_k$, their *disjoint union* is the hierarchy

$$h_1 \cup \dots \cup h_k$$

with node and edge sets

$$V(h_1 \cup \dots \cup h_k) = \bigcup_{\ell=1}^k V(h_\ell)$$

and

$$E(h_1 \cup \dots \cup h_k) = \bigcup_{\ell=1}^k E(h_\ell).$$

The disjoint union is acyclic because each h_ℓ is acyclic and there are no edges between their node sets. When no confusion can arise, we use the same notation for a collection of node-disjoint hierarchies and its disjoint union.

3 Axiomatic framework

A *hierarchy index* is a function $I : \mathcal{H} \rightarrow \mathbb{R}$. For any $h, h' \in \mathcal{H}$, the inequality $I(h') > I(h)$ means that h' is ranked as more hierarchical than h , while $I(h') = I(h)$ means that the two hierarchies are ranked equally.

A hierarchy h' is a *relabeling* of h if there exists a bijection $\phi : V(h) \rightarrow V(h')$ such that, for all $j, i \in V(h)$,

$$(j, i) \in E(h) \Leftrightarrow (\phi(j), \phi(i)) \in E(h').$$

Relabelings have the same directed structure and differ only in their node labels.

Anonymity (A). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **A** if, for all $h, h' \in \mathcal{H}$, $I(h') = I(h)$ whenever h' is a relabeling of h .

A requires the index to depend only on the directed structure of a hierarchy, not on its node labels.

Subordination removal. For each $j \in V(h)$, let

$$P_h(j) = \{p \in V(h) : (p, j) \in E(h)\}$$

denote the set of immediate supervisors of j . For an edge $(j, i) \in E(h)$, define

$$Q_h(j, i) = \left\{ p \in P_h(j) : \begin{array}{l} \text{there is no directed path from } p \text{ to } i \text{ in } h \\ \text{that avoids } j \end{array} \right\}.$$

Definition 1. Let $h \in \mathcal{H}$ and $(j, i) \in E(h)$. The directed graph obtained from h by *removing the subordination relation* (j, i) is the graph h' defined by $V(h') = V(h)$ and

$$E(h') = (E(h) \setminus \{(j, i)\}) \cup \{(p, i) : p \in Q_h(j, i)\}.$$

Thus, the edge from j to i is deleted. Each immediate supervisor p of j is connected directly to i unless p already reaches i along a path that avoids j . If j is a root, then $P_h(j) = \emptyset$, so no edge is added.

The resulting graph is acyclic. Suppose otherwise, and choose a directed cycle in h' with no repeated nodes. Because h is acyclic, the cycle contains an added edge (p, i) . Removing that edge from the cycle leaves a directed path from i to p . This path contains no added edge: every added edge terminates at i , so such an edge would cause the path to revisit its initial node i . The path therefore uses only edges of h . But h also contains the directed path

$$p \rightarrow j \rightarrow i.$$

Together, these paths form a directed cycle in h , a contradiction. Thus, h' is a hierarchy.

An edge $(j, i) \in E(h)$ is *redundant* if h contains a directed path from j to i other than the edge (j, i) itself.

Subordination Removal (SR). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **SR** if, for every $h \in \mathcal{H}$ and every hierarchy h' obtained from h by removing a subordination relation, $I(h) \geq I(h')$. The inequality is strict when the removed edge is non-redundant in h .

SR requires the removal of a reporting relation not to increase measured hierarchy and the removal of a non-redundant edge to decrease it strictly.

A and **SR** are adapted from the ordinal framework of Carbonell-Nicolau (2025a,b). That framework also imposes replication invariance. The next subsection considers replication invariance together with two alternative replication requirements.

3.1 Replication principles

For $h \in \mathcal{H}$ and an integer $k \geq 1$, define the k -fold replication of h , denoted by h^k , as the hierarchy whose node set is

$$V(h^k) = V(h) \times \{1, \dots, k\}$$

and whose edge set is

$$E(h^k) = \{((j, \ell), (i, \ell)) : (j, i) \in E(h), \ell \in \{1, \dots, k\}\}.$$

Thus, h^k is the disjoint union of k tagged copies of h . In particular, h^1 is a relabeling of h , and $|V(h^k)| = k|V(h)|$.

Replication preserves the structure within each copy but changes the size and fragmentation of the population. Its effect on measured hierarchy therefore depends on the question being asked. We consider three alternatives: hierarchy per person, aggregate hierarchy, and concentration of hierarchy.

Hierarchy per person. If the index measures the hierarchical position of a typical individual, replication should not change its value.

Intensive Replication Principle (IRP). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **IRP** if, for every $h \in \mathcal{H}$ and every integer $k \geq 1$, $I(h^k) = I(h)$.

Aggregate hierarchy. If the index measures aggregate hierarchical structure, replication should increase its value whenever the original hierarchy contains at least one subordination relation.

Extensive Replication Principle (ERP). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **ERP** if, for every $h \in \mathcal{H}$ with $E(h) \neq \emptyset$ and every integer $k > 1$, $I(h^k) > I(h)$.

ERP requires only a strict increase. It does not impose linear scaling with the number of copies, which will follow for the depth–span total indices studied below.

Concentration of hierarchy. If the index measures hierarchical structure relative to the number of potential pairs of individuals, replication should reduce its value by creating disconnected copies while increasing the number of potential pairs.

Concentration Replication Principle (CRP). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **CRP** if, for every $h \in \mathcal{H}$ with $E(h) \neq \emptyset$ and every integer $k > 1$, $I(h^k) < I(h)$.

CRP requires only a strict decrease under replication. It does not impose a particular scaling formula. For the pair-normalized concentration indices studied below, replication yields

$$I(h^k) = \frac{|V(h)| - 1}{k|V(h)| - 1} I(h).$$

When $I(h) > 0$ and $k > 1$, the factor is strictly less than one, so $I(h^k) < I(h)$.

No hierarchy index can satisfy more than one of **IRP**, **ERP**, and **CRP** on non-flat hierarchies.

Remark 1 (economic analogues). The three principles have counterparts in economic measurement. Replication invariance parallels the Population Principle in inequality measurement (Dalton, 1920). An increase under replication parallels total measures, such as the aggregate poverty gap or total welfare (Blackorby et al., 2005). A decrease under replication parallels concentration measures, for which reproducing a pattern across independent units lowers its concentration relative to the total population (Hirschman, 1964).

3.2 Unrestricted upward extension

Throughout this and the next two subsections, roots, leaves, and depths refer to the hierarchies before cross-edges are added. A *cross-edge* is an edge whose endpoints belong to different hierarchies. Attachment operators are defined for node-disjoint hierarchies.

All attachment operations defined below preserve acyclicity. The two hierarchies are node-disjoint, and every cross-edge points from the first hierarchy to the second. A directed path that enters the second hierarchy therefore cannot return to the first, so no directed cycle can contain a cross-edge. Any directed cycle would consequently lie within one of the original hierarchies, contradicting its acyclicity.

For node-disjoint hierarchies $h_o, h \in \mathcal{H}$, let

$$h_o \oplus h$$

be the set of hierarchies obtained from the disjoint union of h_o and h by adding edges from leaves of h_o to roots of h , subject to the following requirements:

1. every leaf of h_o has an outgoing edge to at least one root of h ;
2. no other cross-edges are added.

Thus, h_o is the upper hierarchy and h is the lower hierarchy. Not every root of h need receive an edge.

The set $h_o \oplus h$ is nonempty: every finite, nonempty DAG has a root and a leaf, so one may connect every leaf of h_o to any fixed root of h . The set may contain more than one hierarchy because the leaves of h_o can be attached to the roots of h in different ways. The operator is not symmetric: in general,

$$h_o \oplus h \neq h \oplus h_o.$$

Upward Extension (UE). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **UE** if, for all node-disjoint $h_o, h \in \mathcal{H}$ and every $h^* \in h_o \oplus h$, $I(h^*) > I(h)$.

UE requires every admissible upward attachment to raise the index, even when the upper hierarchy is flat (Figure 1). It is therefore unrestricted with respect to the structure of the upper hierarchy.

3.3 Unrestricted downward extension

For node-disjoint hierarchies $h, h_o \in \mathcal{H}$, let

$$h \boxplus h_o$$

be the set of hierarchies obtained from the disjoint union of h and h_o by adding edges from maximum-depth leaves of h to roots of h_o , subject to the following requirements:

1. every root of h_o has an incoming edge from at least one maximum-depth leaf of h ;
2. no other cross-edges are added.

Thus, h is the upper hierarchy and h_o is the lower hierarchy. A maximum-depth leaf has maximum depth among all nodes of h , not merely among the nodes in its connected component. Not every maximum-depth leaf of h need have an outgoing edge to h_o .

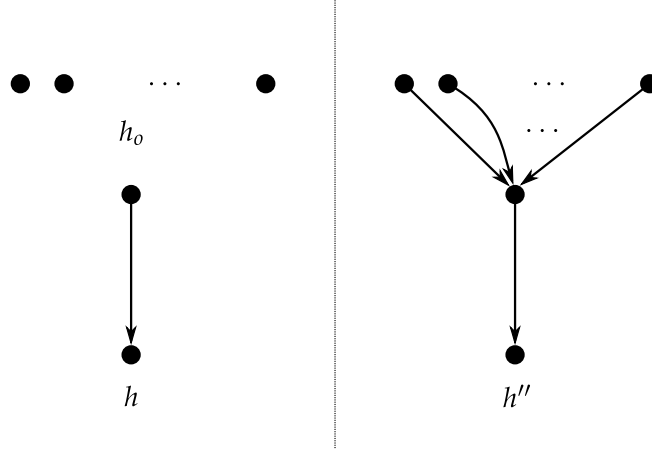


Figure 1: An upward attachment: $h^* \in h_o \oplus h$.

The set $h \boxplus h_o$ is nonempty. Because h is finite, it has a maximum-depth node. Every such node is a leaf: if it had an outgoing edge, the target would have strictly greater depth. Because h_o has a root, one may connect any fixed maximum-depth leaf of h to every root of h_o .

The set $h \boxplus h_o$ may contain more than one hierarchy because the roots of h_o can be attached to the maximum-depth leaves of h in different ways. The operator is not symmetric: in general,

$$h \boxplus h_o \neq h_o \boxplus h.$$

Downward Extension (DE). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **DE** if, for all node-disjoint $h, h_o \in \mathcal{H}$ and every $h^* \in h \boxplus h_o$, $I(h^*) > I(h)$.

DE requires every admissible downward attachment to raise the index, even when the lower hierarchy is flat (**Figure 2**). Requiring attachment at maximum-depth leaves places the new structure at the bottom boundary of h . The condition is unrestricted with respect to the structure of the lower hierarchy.

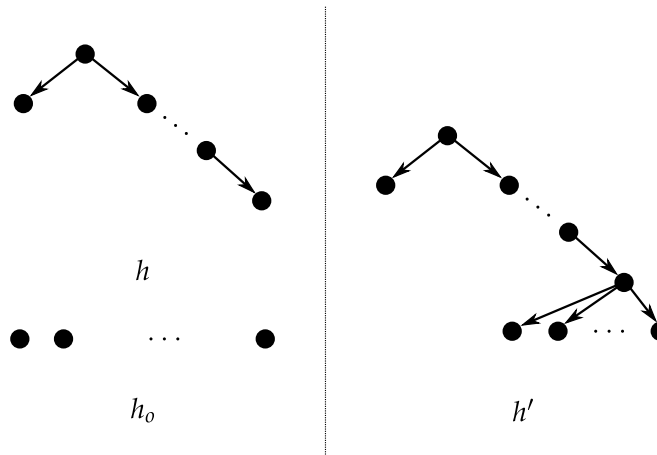


Figure 2: A downward attachment: $h^* \in h \boxplus h_o$.

3.4 Conditional extensions

For node-disjoint hierarchies $h_1, h_2 \in \mathcal{H}$, let

$$h_1 \uplus h_2$$

be the set of hierarchies obtained from their disjoint union by adding a nonempty set of cross-edges, each directed from a leaf of h_1 to a root of h_2 . Thus, h_1 is the upper hierarchy and h_2 is the lower hierarchy.

The attachment may be partial: not every leaf of h_1 need have an outgoing cross-edge, and not every root of h_2 need receive one. The set $h_1 \uplus h_2$ is nonempty: every finite, nonempty DAG has a root and a leaf, so one may add an edge from any leaf of h_1 to any root of h_2 . The set may contain more than one hierarchy. The operator is not symmetric: in general,

$$h_1 \uplus h_2 \neq h_2 \uplus h_1.$$

Moreover, for node-disjoint $h_o, h \in \mathcal{H}$,

$$h_o \oplus h \subseteq h_o \uplus h$$

and

$$h \boxplus h_o \subseteq h \uplus h_o.$$

Thus, $\uplus$ permits upward attachments that do not connect every leaf of the upper hierarchy and downward attachments that need not originate at maximum-depth leaves or reach every root of the lower hierarchy.

Upward Extension* (UE*). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **UE*** if, for all node-disjoint $h_o, h \in \mathcal{H}$ and every $h^* \in h_o \uplus h$,

$$I(h_o) \geq I(h) \Rightarrow I(h^*) > I(h).$$

Downward Extension* (DE*). A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ satisfies **DE*** if, for all node-disjoint $h, h_o \in \mathcal{H}$ and every $h^* \in h \uplus h_o$,

$$I(h_o) \geq I(h) \Rightarrow I(h^*) > I(h).$$

In **UE***, h_o is the upper hierarchy and h is the lower base. In **DE***, h is the upper base and h_o is the lower hierarchy.

In both axioms, the strict increase is required only when the appended hierarchy h_o has an index value at least as high as the base hierarchy h . The operator $\uplus$ permits more attachment patterns than either $\oplus$ or $\boxplus$, while the dominance premise restricts the hierarchy pairs for which an increase is required. The conditional axioms therefore broaden the admissible attachments but narrow the domain of required monotonicity.

4 A class of depth–span indices

This section studies hierarchy measures constructed from individual depth and span. For a hierarchy $h \in \mathcal{H}$ and a node $v \in V(h)$, recall that the depth $s_h(v)$ counts the node’s direct and indirect supervisors, while the span $\sigma_h(v)$ counts its direct and indirect subordinates. A local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \rightarrow \mathbb{R}$ assigns a score to each depth–span pair, and a hierarchy index aggregates these scores using a specified population normalization. The framework therefore separates two choices: how depth and span determine an individual’s structural position and how local scores are aggregated across organizations of different sizes.

The restriction to depth and span is substantive. It treats individuals with the same depth and span as occupying the same hierarchical position, even when the surrounding authority relations differ. A depth–span index is therefore insensitive to structural changes that leave every node’s depth and span unchanged. It does not separately record path lengths, the number of paths between nodes, the division between direct and indirect relations, or the internal organization of a node’s ancestor and descendant sets.

This restriction is appropriate when the numbers of supervisors and subordinates are the primary features of interest. If access to authority depends primarily on the number of supervisors an individual faces, depth measures upward exposure. If influence depends primarily on the number of individuals in a person’s chain of command, span measures downward reach. The restriction is less appropriate when direct supervision, reporting-path length, or the arrangement of intermediate supervisors matters independently of the numbers of supervisors and subordinates.

4.1 Depth–span average indices

We first consider the average population normalization, which evaluates hierarchy per person by averaging individual depth–span scores across the organization.

Definition 2. A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ is a *depth–span average index* if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \rightarrow \mathbb{R}$ such that, for every $h \in \mathcal{H}$,

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

Such a function g is called a *depth–span representation* of I .

Every depth–span average index satisfies **A**, because relabeling preserves every depth–span pair. It also satisfies **IRP**, because replication reproduces the same multiset of depth–span pairs in each copy and therefore leaves their average unchanged.

While this normalization captures hierarchy per person, its combination with the depth–span restriction creates a sharp tension between unrestricted upward extension and strict responsiveness to the removal of a non-redundant subordination relation. The following results make this incompatibility precise.

Lemma 1. *If a depth–span average index I satisfies **UE**, then it is bounded above on $\mathcal{H}$.*

Proof. Suppose, toward a contradiction, that I is unbounded above. Fix a depth–span representation

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

Let e denote a one-node hierarchy, let P_2 denote a two-node directed chain, and define

$$c_0 = g(0, 0) = I(e)$$

and

$$\Delta = g(0, 1) + g(1, 0) - 2c_0 = 2[I(P_2) - I(e)].$$

Because I is unbounded above, choose $C \in \mathcal{H}$ such that

$$I(C) > c_0 + \Delta + \max\{\Delta, 0\}.$$

Write $q = |V(C)|$. Let $r \notin V(C)$ and define $h = C \cup \{r\}$, where r is isolated. By the depth–span average representation,

$$I(h) = \frac{qI(C) + c_0}{q + 1}.$$

Let $h_o = \{v^*\}$ be a one-node hierarchy, and form $h^* \in h_o \oplus h$ by adding the edge $v^* \rightarrow r$. This is an admissible upward attachment. The hierarchy C is unchanged. Before the attachment, both v^* and r have depth–span pair $(0, 0)$. After the attachment, v^* has pair $(0, 1)$ and r has pair $(1, 0)$. Therefore,

$$I(h^*) = \frac{qI(C) + g(0, 1) + g(1, 0)}{q + 2} = \frac{qI(C) + 2c_0 + \Delta}{q + 2}.$$

It follows that

$$I(h^*) - I(h) = \frac{(q + 1)[qI(C) + 2c_0 + \Delta] - (q + 2)[qI(C) + c_0]}{(q + 1)(q + 2)} = \frac{(q + 1)\Delta - q[I(C) - c_0]}{(q + 1)(q + 2)}.$$

Our choice of C gives

$$I(C) - c_0 > \Delta + \max\{\Delta, 0\}.$$

Hence,

$$q[I(C) - c_0] - (q + 1)\Delta > q[\Delta + \max\{\Delta, 0\}] - (q + 1)\Delta = q \max\{\Delta, 0\} - \Delta \geq 0.$$

Thus,

$$(q + 1)\Delta - q[I(C) - c_0] < 0.$$

Therefore, $I(h^*) < I(h)$, contradicting **UE**. ■

For positive integers n and m , let $K_{n,m}$ denote the complete two-layer hierarchy with n roots and m leaves, with an edge from every root to every leaf and no other edges. For positive integers a, b, c , let $K_{a,b,c}$ denote the complete three-layer hierarchy with layers of sizes a, b , and c , respectively, with an edge from every node in the first layer to every node in the second, an edge from every node in the second layer to every node in the third, and no other edges.

The next result strengthens **Lemma 1**. **Lemma 1** shows that **UE** forces every depth–span average index to be bounded above. The proof below shows that, together with the depth–span restriction, this boundedness forces a strong degeneracy in the local representation. Complete two-layer attachments imply that the index admits an equivalent depth–span representation in which every root with positive span receives the supremum value $M = \sup_{h \in \mathcal{H}} I(h)$. Complete three-layer attachments then imply that every node with positive span receives the same value. Thus, in this equivalent

representation, once a node has at least one subordinate, its local score no longer depends on its depth. The final step applies this conclusion to a diamond hierarchy. Removing a non-redundant edge changes the depth–span pairs of some non-leaf nodes, but those nodes remain non-leaves and therefore retain local score M . The unique leaf has the same depth–span pair before and after the removal. Hence, the average score is unchanged, even though **SR** requires a strict decrease when a non-redundant subordination relation is removed.

Theorem 1. *No depth–span average index satisfies both **UE** and **SR**.*

Proof. Suppose, toward a contradiction, that a depth–span average index I satisfies **UE** and **SR**. Fix a depth–span representation

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

By **Lemma 1**, **UE** implies that I is bounded above. Define

$$M = \sup_{h \in \mathcal{H}} I(h) \quad \text{and} \quad c_0 = I(e) = g(0, 0),$$

where e denotes a one-node hierarchy. In particular, M is finite.

Choose a sequence $(C_k)_{k \geq 1}$ such that $I(C_k) \rightarrow M$. For each k , choose a positive integer r_k such that $|V(C_k^{r_k})| \geq k$. Replication invariance gives $I(C_k^{r_k}) = I(C_k)$. Replacing each C_k by $C_k^{r_k}$ and retaining the notation C_k , we may therefore assume that

$$I(C_k) \rightarrow M \quad \text{and} \quad |V(C_k)| \rightarrow \infty.$$

Taking node-disjoint copies throughout, let F_m be a flat hierarchy with m isolated nodes and define

$$h_{k,m} = C_k \cup F_m.$$

Then

$$I(h_{k,m}) = \frac{|V(C_k)|I(C_k) + mc_0}{|V(C_k)| + m}.$$

Let F_n be a flat hierarchy with n isolated nodes that is node-disjoint from $h_{k,m}$. Form an upward attachment

$$h_{k,m,n}^* \in F_n \oplus h_{k,m}$$

by adding an edge from every node of F_n to every node of F_m . This is admissible because every node of F_n is a leaf and has an outgoing edge to a root of the base. The definition of **UE** does not require every root of the base to receive a cross-edge.

The subgraph induced by $V(C_k)$ remains unchanged, and no cross-edge is incident to a node of C_k , while the subgraph on $V(F_n) \cup V(F_m)$ becomes $K_{n,m}$. Thus,

$$I(h_{k,m,n}^*) = \frac{|V(C_k)|I(C_k) + (n+m)I(K_{n,m})}{|V(C_k)| + n + m}.$$

By **UE**, $I(h_{k,m,n}^*) > I(h_{k,m})$. Write $q_k = |V(C_k)|$. Cross-multiplication gives

$$q_k [(n+m)I(K_{n,m}) - nI(C_k) - mc_0] + m(n+m)[I(K_{n,m}) - c_0] > 0.$$

Dividing by q_k , using $q_k \rightarrow \infty$ and $I(C_k) \rightarrow M$, and taking limits yields

$$(n + m)I(K_{n,m}) \geq nM + mc_0.$$

Hence,

$$I(K_{n,m}) \geq \frac{nM + mc_0}{n + m}. \quad (1)$$

In $K_{n,m}$, each root has depth zero and span m , while each leaf has depth n and span zero. Therefore,

$$I(K_{n,m}) = \frac{ng(0, m) + mg(n, 0)}{n + m}.$$

Combining this identity with (1) yields

$$ng(0, m) + mg(n, 0) \geq nM + mc_0. \quad (2)$$

Fix $m \geq 1$. As $n \rightarrow \infty$, the right-hand side of (1) converges to M . Since $I(K_{n,m}) \leq M$, it follows that $I(K_{n,m}) \rightarrow M$. Using

$$I(K_{n,m}) = \frac{ng(0, m) + mg(n, 0)}{n + m},$$

we have

$$\frac{g(n, 0)}{n} = \frac{1}{m} \left[\left(1 + \frac{m}{n}\right) I(K_{n,m}) - g(0, m) \right].$$

Therefore,

$$\frac{g(n, 0)}{n} \rightarrow \frac{M - g(0, m)}{m}.$$

Because the same sequence $g(n, 0)/n$ converges to this expression for every fixed $m \geq 1$, the expression $[M - g(0, m)]/m$ is independent of m . Therefore, there exists a constant $\theta \in \mathbb{R}$ such that

$$\theta = \lim_{n \rightarrow \infty} \frac{g(n, 0)}{n}$$

and $g(0, m) = M - m\theta$ for every $m \geq 1$.

Define

$$\widehat{g}(s, \sigma) = g(s, \sigma) + \theta(\sigma - s).$$

For every hierarchy h ,

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

because each ordered ancestor–descendant pair contributes once to the depth of its lower node and once to the span of its upper node. Consequently,

$$\sum_{v \in V(h)} \theta[\sigma_h(v) - s_h(v)] = 0,$$

so $\widehat{g}$ represents the same index as g . Moreover, $\widehat{g}(0, 0) = g(0, 0) = c_0$, and $\widehat{g}(0, m) = M$ for every $m \geq 1$.

We next bound the evaluations of leaves under this equivalent representation. Since (2) is invariant under the change from g to $\widehat{g}$, it gives

$$n\widehat{g}(0, m) + m\widehat{g}(n, 0) \geq nM + mc_0.$$

Using $\widehat{g}(0, m) = M$, we obtain $\widehat{g}(n, 0) \geq c_0$. On the other hand,

$$I(K_{n,m}) = \frac{nM + m\widehat{g}(n, 0)}{n + m} \leq M,$$

which implies $\widehat{g}(n, 0) \leq M$. Thus,

$$c_0 \leq \widehat{g}(n, 0) \leq M \quad (3)$$

for every $n \geq 1$.

Taking node-disjoint copies, start from the base $C_k \cup K_{b,c}$, and append a flat hierarchy of a nodes above $K_{b,c}$ by connecting every appended node to every root of $K_{b,c}$. This is admissible because every appended node is a leaf and has an outgoing edge to a root of the base. The definition of **UE** does not require every root of the base to receive a cross-edge.

The subgraph induced by $V(C_k)$ remains unchanged, while the subgraph on the remaining nodes becomes $K_{a,b,c}$. Applying **UE**, cross-multiplying, dividing by $|V(C_k)|$, using $I(C_k) \rightarrow M$ and $|V(C_k)| \rightarrow \infty$, and taking limits yields

$$(a + b + c)I(K_{a,b,c}) \geq aM + (b + c)I(K_{b,c}). \quad (4)$$

In $K_{b,c}$, the roots have coordinates $(0, c)$ and the leaves have coordinates $(b, 0)$. Hence,

$$(b + c)I(K_{b,c}) = b\widehat{g}(0, c) + c\widehat{g}(b, 0) = bM + c\widehat{g}(b, 0).$$

In $K_{a,b,c}$, each root has coordinates $(0, b + c)$, each middle node has coordinates (a, c) , and each leaf has coordinates $(a + b, 0)$. Because depth and span count indirect as well as direct relations, each root has all c leaves as descendants, and each leaf has all a roots as ancestors. Therefore,

$$(a + b + c)I(K_{a,b,c}) = aM + b\widehat{g}(a, c) + c\widehat{g}(a + b, 0).$$

Substituting these identities into (4) gives

$$aM + b\widehat{g}(a, c) + c\widehat{g}(a + b, 0) \geq aM + bM + c\widehat{g}(b, 0).$$

Thus,

$$\widehat{g}(a, c) \geq M + \frac{c}{b} [\widehat{g}(b, 0) - \widehat{g}(a + b, 0)]. \quad (5)$$

Since $I(K_{a,b,c}) \leq M$, we also have

$$aM + b\widehat{g}(a, c) + c\widehat{g}(a + b, 0) \leq (a + b + c)M.$$

Hence,

$$\widehat{g}(a, c) \leq M + \frac{c}{b} [M - \widehat{g}(a + b, 0)]. \quad (6)$$

Fix $a, c \geq 1$ and let $b \rightarrow \infty$. By (3), both $\widehat{g}(b, 0)$ and $\widehat{g}(a + b, 0)$ lie in $[c_0, M]$. Therefore, the bracketed expressions in (5) and (6) are uniformly bounded. Since $c/b \rightarrow 0$, both bounds converge

to M . For every $b \geq 1$, $\widehat{g}(a, c)$ lies between these bounds. Because $\widehat{g}(a, c)$ does not depend on b , it follows that $\widehat{g}(a, c) = M$.

Combining this conclusion with $\widehat{g}(0, c) = M$ for every $c \geq 1$, we obtain

$$\widehat{g}(s, \sigma) = M \quad (7)$$

for every $s \geq 0$ and $\sigma \geq 1$.

Finally, consider the four-node diamond hierarchy h with edges

$$x \rightarrow y, \quad x \rightarrow z, \quad y \rightarrow w, \quad z \rightarrow w.$$

The edge $x \rightarrow y$ is non-redundant because there is no other directed path from x to y . The depth–span coordinates of h are

$$x : (0, 3), \quad y : (1, 1), \quad z : (1, 1), \quad w : (3, 0).$$

By (7),

$$I(h) = \frac{3M + \widehat{g}(3, 0)}{4}.$$

Remove the non-redundant edge $x \rightarrow y$. Since x is a root, the subordination-removal operation adds no bypass edge. The resulting hierarchy h' has edges

$$x \rightarrow z, \quad y \rightarrow w, \quad z \rightarrow w.$$

Its depth–span coordinates are

$$x : (0, 2), \quad y : (0, 1), \quad z : (1, 1), \quad w : (3, 0).$$

The depth of w remains three because its ancestor set remains $\{x, y, z\}$: node x continues to reach w through z . Again by (7),

$$I(h') = \frac{3M + \widehat{g}(3, 0)}{4} = I(h).$$

This contradicts the strict inequality required by **SR** when a non-redundant subordination relation is removed. ■

The incompatibility is driven by the interaction of averaging and the depth–span restriction. Satisfaction of **UE** forces the index to admit an equivalent representation that assigns the supremum score to every non-leaf position. The index then cannot respond strictly to a non-redundant edge removal that preserves the depth–span pair of the unique leaf and leaves every other node as a non-leaf.

The preceding impossibility result isolates a specific limitation. Under **UE**, a large flat upper population may attach to an isolated root of a disconnected base, leaving the high-valued subgraph of the base unchanged while diluting its population average.

Within the depth–span average framework, the unconditional impossibility established above is specific to **UE**: the remaining extension axioms are jointly compatible with strict subordination removal under the conditions below. The conditional extension axioms impose a dominance premise on the appended hierarchy, while **DE** restricts attachment to the maximum-depth boundary. The next result identifies a broad class of depth–span average indices satisfying **SR**, **UE***, and **DE***, together with a sufficient condition for **DE**.

Theorem 2. Let I be a depth–span average index with representation

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

Suppose that g is weakly increasing in both coordinates:

$$g(s+1, \sigma) \geq g(s, \sigma) \quad \text{and} \quad g(s, \sigma+1) \geq g(s, \sigma)$$

for every $s, \sigma \in \mathbb{Z}_+$. Suppose, moreover, that at least one of the following uniform strictness conditions holds:

$$g(s+1, \sigma) > g(s, \sigma) \quad \text{for every } s, \sigma \in \mathbb{Z}_+, \quad (\text{SD})$$

or

$$g(s, \sigma+1) > g(s, \sigma) \quad \text{for every } s, \sigma \in \mathbb{Z}_+. \quad (\text{SS})$$

Then I satisfies **SR**, **UE***, and **DE***.

If, in addition, I admits a depth–span representation $\widehat{g}$ that is weakly increasing in both coordinates and satisfies

$$\widehat{g}(D(h)+1, 0) > I(h), \quad \text{for every } h \in \mathcal{H}, \quad (\text{DB})$$

where

$$D(h) = \max_{v \in V(h)} s_h(v),$$

then I also satisfies **DE**.

Proof. We first establish **SR**. Consider a hierarchy $h \in \mathcal{H}$, an edge $(j, i) \in E(h)$, and the hierarchy h' obtained by removing the corresponding subordination relation.

Every bypass edge added in the subordination-removal operation replaces a directed path already present in h . Hence, the operation creates no new ancestor–descendant pair. It follows that

$$s_{h'}(v) \leq s_h(v) \quad \text{and} \quad \sigma_{h'}(v) \leq \sigma_h(v)$$

for every $v \in V(h)$. By weak coordinatewise monotonicity, no local score increases. Since the population is unchanged, this implies $I(h') \leq I(h)$.

Suppose now that (j, i) is non-redundant. Then h contains no directed path from j to i other than the removed edge. No added bypass edge can restore a path from j to i , because every such edge originates at an immediate supervisor of j , which j cannot reach in an acyclic hierarchy. Therefore, j no longer reaches i in h' . In particular,

$$s_{h'}(i) < s_h(i) \quad \text{and} \quad \sigma_{h'}(j) < \sigma_h(j).$$

If **(SD)** holds, the strict decrease in the depth of i , together with the weak decrease in its span, strictly lowers its local score. If **(SS)** holds, the strict decrease in the span of j , together with the weak decrease in its depth, strictly lowers its local score. In either case, at least one local score decreases strictly, while no local score increases. Hence, $I(h') < I(h)$. Thus, I satisfies **SR**.

We next prove **UE***. Let $h_o, h \in \mathcal{H}$ be node-disjoint hierarchies such that $I(h_o) \geq I(h)$, and take $h^* \in h_o \uplus h$. The attachment adds at least one cross-edge and removes no ancestor–descendant relation. Therefore,

$$s_{h^*}(v) \geq s_{h_o \cup h}(v) \quad \text{and} \quad \sigma_{h^*}(v) \geq \sigma_{h_o \cup h}(v)$$

for every $v \in V(h_o) \cup V(h)$.

Every cross-edge increases the span of its source and the depth of its target. If **(SD)** holds, repeated application of **(SD)** shows that the score of at least one target rises strictly. If **(SS)** holds, repeated application of **(SS)** shows that the score of at least one source rises strictly. Thus, under either condition, at least one local score increases strictly while none decreases.

For every hierarchy $\tilde{h} \in \mathcal{H}$, define

$$S(\tilde{h}) = \sum_{v \in V(\tilde{h})} g(s_{\tilde{h}}(v), \sigma_{\tilde{h}}(v)).$$

We therefore have

$$S(h^*) > S(h_o) + S(h).$$

Consequently,

$$I(h^*) = \frac{S(h^*)}{|V(h_o)| + |V(h)|} > \frac{|V(h_o)|I(h_o) + |V(h)|I(h)}{|V(h_o)| + |V(h)|} \geq I(h).$$

Thus, I satisfies **UE***.

The proof of **DE*** is identical. Let $h, h_o \in \mathcal{H}$ be node-disjoint, suppose that $I(h_o) \geq I(h)$, and take $h^* \in h \uplus h_o$. Every cross-edge increases the span of its source and the depth of its target, while decreasing no coordinate. Under either **(SD)** or **(SS)**, at least one local score increases strictly and none decreases. Therefore,

$$S(h^*) > S(h) + S(h_o).$$

The same weighted-average argument yields $I(h^*) > I(h)$. Thus, I satisfies **DE***.

Finally, suppose that I admits a depth–span representation $\hat{g}$ that is weakly increasing in both coordinates and satisfies **(DB)**. Let $h, h_o \in \mathcal{H}$ be node-disjoint, and take $h^* \in h \boxplus h_o$. Write $D = D(h)$.

Every root of h_o receives an incoming edge from a maximum-depth leaf of h . Therefore, each root of h_o has at least $D + 1$ ancestors in h^* . Every node of the finite DAG h_o is reachable from one of its roots. Therefore, every non-root node of h_o is reachable from a root that receives an incoming edge from a maximum-depth leaf of h , so it also has at least $D + 1$ ancestors in h^* . Hence, $s_{h^*}(w) \geq D + 1$ for every $w \in V(h_o)$.

By weak coordinatewise monotonicity of $\hat{g}$ and **(DB)**,

$$\hat{g}(s_{h^*}(w), \sigma_{h^*}(w)) \geq \hat{g}(D + 1, 0) > I(h)$$

for every $w \in V(h_o)$.

Write

$$n = |V(h)| \quad \text{and} \quad n_o = |V(h_o)|.$$

The aggregate score of the appended nodes therefore satisfies

$$\sum_{w \in V(h_o)} \hat{g}(s_{h^*}(w), \sigma_{h^*}(w)) > n_o I(h).$$

The attachment creates new ancestor–descendant relations but removes none. Thus, no depth or span of any original node decreases. Weak coordinatewise monotonicity of $\widehat{g}$ gives

$$\sum_{v \in V(h)} \widehat{g}(s_{h^*}(v), \sigma_{h^*}(v)) \geq nI(h).$$

Adding these inequalities yields

$$\sum_{v \in V(h^*)} \widehat{g}(s_{h^*}(v), \sigma_{h^*}(v)) > (n + n_o)I(h).$$

Since $\widehat{g}$ represents I and $|V(h^*)| = n + n_o$, we conclude that $I(h^*) > I(h)$. Thus, I satisfies **DE**. ■

Remark 2. Weak coordinatewise monotonicity does not by itself guarantee **SR**. In particular, the uniform strictness requirement in **Theorem 2** cannot simply be omitted while retaining its conclusion concerning **SR**.

Consider the depth–span representation

$$g(s, \sigma) = \begin{cases} s, & \text{if } \sigma = 0, \\ \max\{s, 1\}, & \text{if } \sigma \geq 1, \end{cases}$$

and define

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

The function g is weakly increasing in both coordinates. It does not, however, satisfy uniform strictness in depth, because

$$g(1, 1) = g(0, 1) = 1,$$

and it does not satisfy uniform strictness in span, because

$$g(1, 1) = g(1, 0) = 1.$$

Moreover, the downward-boundary condition holds. To see this, let $h \in \mathcal{H}$ and write $D = D(h)$. If $D = 0$, then h is flat, so

$$I(h) = g(0, 0) = 0 < 1 = g(1, 0).$$

If $D \geq 1$, then every node $v \in V(h)$ satisfies

$$g(s_h(v), \sigma_h(v)) \leq D.$$

Indeed, if $\sigma_h(v) = 0$, then its score is $s_h(v) \leq D$. If $\sigma_h(v) \geq 1$, then its score is

$$\max\{s_h(v), 1\} \leq D.$$

Consequently,

$$I(h) \leq D < D + 1 = g(D + 1, 0).$$

Thus, the displayed representation g is weakly increasing in both coordinates and satisfies

$$g(D(h) + 1, 0) > I(h)$$

for every $h \in \mathcal{H}$. Hence, it satisfies (DB).

Nevertheless, the index violates the strict part of SR. Consider the four-node diamond hierarchy h with edges

$$a \rightarrow b, \quad a \rightarrow c, \quad b \rightarrow d, \quad c \rightarrow d.$$

Its depth–span coordinates are

$$a : (0, 3), \quad b : (1, 1), \quad c : (1, 1), \quad d : (3, 0).$$

The corresponding scores are

$$g(0, 3) = 1, \quad g(1, 1) = 1, \quad g(1, 1) = 1, \quad g(3, 0) = 3.$$

Hence,

$$I(h) = \frac{1 + 1 + 1 + 3}{4} = \frac{3}{2}.$$

Remove the non-redundant edge $a \rightarrow b$. Since a is a root, the subordination-removal operation adds no bypass edge. The resulting hierarchy h' has edges

$$a \rightarrow c, \quad b \rightarrow d, \quad c \rightarrow d,$$

and depth–span coordinates

$$a : (0, 2), \quad b : (0, 1), \quad c : (1, 1), \quad d : (3, 0).$$

The corresponding scores are again

$$g(0, 2) = 1, \quad g(0, 1) = 1, \quad g(1, 1) = 1, \quad g(3, 0) = 3.$$

Therefore,

$$I(h') = \frac{3}{2} = I(h).$$

The removal of a non-redundant edge fails to lower the index strictly, so I violates SR.

This example isolates the role of uniform strictness in the subordination-removal conclusion of Theorem 2. Weak coordinatewise monotonicity and (DB) may both hold, yet the strict part of SR may fail.

Remark 3. Uniform strictness does not by itself guarantee DE. In particular, the condition (DB) in Theorem 2 cannot simply be omitted while retaining its conclusion concerning unrestricted downward extension.

Consider the depth–span representation

$$g(s, \sigma) = s + 8(1 - 2^{-\sigma}),$$

and define

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

The function g is uniformly strictly increasing in both coordinates. Indeed,

$$g(s+1, \sigma) - g(s, \sigma) = 1 > 0$$

and

$$g(s, \sigma+1) - g(s, \sigma) = 8 \left(2^{-\sigma} - 2^{-(\sigma+1)} \right) = 2^{2-\sigma} > 0.$$

Consequently, the index satisfies **SR**, **UE***, and **DE*** by **Theorem 2**.

The displayed representation nevertheless fails (**DB**) on a three-node hierarchy. Let h be the out-star with edges

$$a \rightarrow b \quad \text{and} \quad a \rightarrow c.$$

Its depth–span coordinates are

$$a : (0, 2), \quad b : (1, 0), \quad c : (1, 0).$$

Therefore,

$$g(0, 2) = 8 \left(1 - \frac{1}{4} \right) = 6$$

and $g(1, 0) = 1$. It follows that

$$I(h) = \frac{6+1+1}{3} = \frac{8}{3}.$$

Since $D(h) = 1$, we have

$$g(D(h) + 1, 0) = g(2, 0) = 2.$$

Hence,

$$g(D(h) + 1, 0) = 2 < \frac{8}{3} = I(h).$$

Thus, failure of (**DB**) for the displayed representation is already detected by this three-node out-star.

The index also violates **DE**. For each positive integer m , let h_o^m be a flat hierarchy consisting of m isolated nodes, taking a node-disjoint copy if necessary. Form $h_m^* \in h \boxplus h_o^m$ by adding an edge from b to every node of h_o^m . This is an admissible downward attachment because b is a maximum-depth leaf of h , and every root of h_o^m receives an incoming edge from b .

In h_m^* , the original nodes have depth–span coordinates

$$a : (0, m+2), \quad b : (1, m), \quad c : (1, 0),$$

while every appended node has coordinates $(2, 0)$. Therefore,

$$I(h_m^*) = \frac{1}{m+3} \left[8(1 - 2^{-(m+2)}) + 1 + 8(1 - 2^{-m}) + 1 + 2m \right].$$

Equivalently,

$$I(h_m^*) = \frac{2m + 18 - 8 \cdot 2^{-(m+2)} - 8 \cdot 2^{-m}}{m+3}.$$

It follows that $I(h_m^*) \rightarrow 2$ as $m \rightarrow \infty$. Since

$$2 < \frac{8}{3} = I(h),$$

we have $I(h_m^*) < I(h)$ for every sufficiently large m . Hence, I violates **DE**. By the contrapositive of **Theorem 2**, I therefore admits no depth–span representation that is weakly increasing in both coordinates and satisfies **(DB)**.

This example isolates the role of **(DB)**. Even uniform strictness in both coordinates does not prevent a large, low-scoring appended population from diluting the base average. The boundary condition in **Theorem 2** rules out this dilution mechanism.

Remark 4 (admissible local evaluations). The conditions in **Theorem 2** restrict the direction, but not the magnitude, of changes in local scores. Weak coordinatewise monotonicity requires additional depth or span not to lower a node’s evaluation, while uniform strictness in at least one coordinate ensures a strict response to the structural changes used in **SR**, **UE***, and **DE***.

These conditions permit sublinear, linear, and superlinear specifications. For example, the functions

$$\begin{aligned} g_1(s, \sigma) &= \sqrt{s} + \sqrt{\sigma}, \\ g_2(s, \sigma) &= as + b\sigma, \quad a, b > 0, \\ g_3(s, \sigma) &= s^2 + \sigma^2 + s + \sigma \end{aligned}$$

are uniformly strictly increasing in both coordinates. The theorem therefore imposes no uniform positive lower bound on score increments and permits diminishing, constant, or increasing increments.

Condition **(DB)** is separate. The preceding examples satisfy the monotonicity and strictness hypotheses, but these properties alone do not establish **DE**. To establish **DE** through **Theorem 2**, one must also exhibit a weakly increasing equivalent representation $\widehat{g}$ such that

$$\widehat{g}(D(h) + 1, 0) > I(h)$$

for every hierarchy h .

The linear specification

$$g_2(s, \sigma) = as + b\sigma, \quad a, b > 0,$$

satisfies this condition through an equivalent representation. Indeed, since every hierarchy satisfies

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

the index represented by g_2 is also represented by

$$\widehat{g}_2(s, \sigma) = (a + b)s.$$

For this representation,

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} (a + b)s_h(v) \leq (a + b)D(h) < (a + b)(D(h) + 1) = \widehat{g}_2(D(h) + 1, 0).$$

Thus, by **Theorem 2**, the linear specification satisfies **DE**.

The sublinear specification

$$g_1(s, \sigma) = \sqrt{s} + \sqrt{\sigma}$$

fails **(DB)** for the displayed representation. Consider the four-node hierarchy h with edges

$$x \rightarrow y, \quad y \rightarrow z_1, \quad y \rightarrow z_2.$$

Its depth–span coordinates are

$$x : (0, 3), \quad y : (1, 2), \quad z_1 : (2, 0), \quad z_2 : (2, 0).$$

Hence, $D(h) = 2$ and

$$I(h) = \frac{\sqrt{3} + 1 + \sqrt{2} + 2\sqrt{2}}{4} = \frac{\sqrt{3} + 1 + 3\sqrt{2}}{4} > \sqrt{3} = g_1(D(h) + 1, 0).$$

Thus, (DB) fails for the displayed representation.

The index also violates DE. For each positive integer m , let F_m be a flat hierarchy with m isolated nodes that is node-disjoint from h . Form $h_m^* \in h \boxplus F_m$ by adding an edge from z_1 to every node of F_m . This is an admissible downward attachment because z_1 is a maximum-depth leaf of h , and every root of F_m receives an incoming edge from z_1 . In h_m^* , the original nodes have coordinates

$$x : (0, m + 3), \quad y : (1, m + 2), \quad z_1 : (2, m), \quad z_2 : (2, 0),$$

while every appended node has coordinates $(3, 0)$. Therefore,

$$I(h_m^*) = \frac{\sqrt{m+3} + 1 + \sqrt{m+2} + \sqrt{2} + \sqrt{m} + \sqrt{2} + m\sqrt{3}}{m+4}.$$

It follows that $I(h_m^*) \rightarrow \sqrt{3}$ as $m \rightarrow \infty$. Since $\sqrt{3} < I(h)$, we have $I(h_m^*) < I(h)$ for every sufficiently large m . Thus, the index violates DE. By the contrapositive of Theorem 2, this index admits no weakly increasing equivalent representation satisfying (DB).

The superlinear specification

$$g_3(s, \sigma) = s^2 + \sigma^2 + s + \sigma$$

fails (DB) for the displayed representation and violates DE. First, the boundary condition fails on large out-stars. Let C_m be the out-star with one root and m leaves. Then $D(C_m) = 1$, the root has coordinates $(0, m)$, each leaf has coordinates $(1, 0)$, and

$$I(C_m) = \frac{m^2 + m + 2m}{m+1} = \frac{m^2 + 3m}{m+1}.$$

On the other hand,

$$g_3(D(C_m) + 1, 0) = g_3(2, 0) = 6.$$

For all sufficiently large m ,

$$I(C_m) > 6 = g_3(D(C_m) + 1, 0),$$

so (DB) fails for the displayed representation.

To see that DE also fails, let B be the two-node chain $u \rightarrow v$, taking B node-disjoint from C_m . The depth–span coordinates in B are

$$u : (0, 1), \quad v : (1, 0),$$

and the sum of its local scores is

$$g_3(0, 1) + g_3(1, 0) = 2 + 2 = 4.$$

Define $h_m = C_m \cup B$. Then $D(h_m) = 1$, and the sum of its local scores is

$$m^2 + 3m + 4.$$

Let $t \notin V(h_m)$ and form $h_m^* \in h_m \boxplus \{t\}$ by adding the edge $v \rightarrow t$. This is an admissible downward attachment because v is a maximum-depth leaf of h_m . The subgraph induced by $V(C_m)$ is unchanged, while the subgraph on $V(B) \cup \{t\}$ becomes a three-node chain with coordinates

$$u : (0, 2), \quad v : (1, 1), \quad t : (2, 0).$$

The sum of its local scores is

$$g_3(0, 2) + g_3(1, 1) + g_3(2, 0) = 6 + 4 + 6 = 16.$$

Thus,

$$I(h_m) = \frac{m^2 + 3m + 4}{m + 3} \quad \text{and} \quad I(h_m^*) = \frac{m^2 + 3m + 16}{m + 4}.$$

The inequality $I(h_m^*) < I(h_m)$ is equivalent to

$$12(m + 3) < m^2 + 3m + 4,$$

or

$$m^2 - 9m - 32 > 0,$$

which holds for all sufficiently large m . Hence, the index violates **DE**. By the contrapositive of **Theorem 2**, this index admits no weakly increasing equivalent representation satisfying **(DB)**.

The preceding calculations show that **(DB)** must be checked separately from the monotonicity and uniform-strictness hypotheses. The linear specification satisfies the boundary condition through an equivalent representation, whereas the square-root and quadratic specifications violate unrestricted downward extension. Hence, by the contrapositive of **Theorem 2**, neither index admits a weakly increasing equivalent representation satisfying **(DB)**.

Remark 5 (equivalent depth–span representations). A depth–span representation need not be unique. For every hierarchy h ,

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

because each ordered ancestor–descendant pair contributes once to the depth of its lower node and once to the span of its upper node. Consequently, if g represents a depth–span average index I , then, for every $\beta > 0$, $\gamma \in \mathbb{R}$, and $\theta \in \mathbb{R}$, the function

$$\tilde{g}(s, \sigma) = \beta g(s, \sigma) + \gamma + \theta(\sigma - s)$$

represents the positive affine transformation $\tilde{I}(h) = \beta I(h) + \gamma$. Indeed,

$$\frac{1}{|V(h)|} \sum_{v \in V(h)} \tilde{g}(s_h(v), \sigma_h(v)) = \beta I(h) + \gamma + \frac{\theta}{|V(h)|} \sum_{v \in V(h)} (\sigma_h(v) - s_h(v)) = \beta I(h) + \gamma.$$

Such a transformation preserves the ranking of hierarchies and therefore preserves the ordinal content of the axioms considered here. In particular, when $\beta = 1$ and $\gamma = 0$, $\tilde{g}$ represents the same index as g .

This identity is especially useful for unilateral specifications. If

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} f(s_h(v)),$$

then I is already a depth–span average index with representation $g(s, \sigma) = f(s)$. If f is weakly increasing and satisfies

$$f(s+1) > f(s), \quad \text{for every } s \in \mathbb{Z}_+,$$

then this representation is weakly increasing in both coordinates and uniformly strictly increasing in depth. It therefore satisfies the monotonicity hypotheses of [Theorem 2](#); strict dependence on span is not required.

A representation of a positive scalar transformation of the index with strict dependence on both coordinates may sometimes be obtained by setting

$$\widehat{g}(s, \sigma) = \beta f(s) + \theta(\sigma - s), \quad \beta > 0, \quad \theta > 0.$$

This function represents βI . Its discrete marginals are

$$\widehat{g}(s, \sigma + 1) - \widehat{g}(s, \sigma) = \theta$$

and

$$\widehat{g}(s+1, \sigma) - \widehat{g}(s, \sigma) = \beta(f(s+1) - f(s)) - \theta.$$

Hence, $\widehat{g}$ is uniformly strictly increasing in both coordinates whenever

$$0 < \theta < \beta \inf_{s \in \mathbb{Z}_+} (f(s+1) - f(s)).$$

When $\beta = 1$, $\widehat{g}$ represents the same index as g . For average depth, $f(s) = s$, and choosing $\beta = 1$ and $\theta = \frac{1}{2}$ gives $\widehat{g}(s, \sigma) = \frac{1}{2}(s + \sigma)$.

The sufficient conditions in [Theorem 2](#) are conditions on an exhibited representation. Adding the zero-sum term $\theta(\sigma - s)$ preserves the index but may change the coordinatewise monotonicity and strictness properties of the local evaluation function. Multiplication by $\beta > 0$ instead produces the positive scalar transformation βI .

[Table 1](#) summarizes the axiomatic compliance of the depth–span average index family established by these results. The fundamental tension within this class is between intensive population normalization and unrestricted vertical extensions. While the population average satisfies [A](#) and [IRP](#), it conflicts with [UE](#), as a large flat upper appendage can arbitrarily dilute the base hierarchy’s average. Nevertheless, the remaining monotonicity properties are compatible with this framework. Weak coordinatewise monotonicity and uniform strictness in at least one coordinate guarantee [SR](#), [UE*](#), and [DE*](#), while [DE](#) follows under the additional representation-level condition ([DB](#)).

The average depth index studied by [Carbonell-Nicolau \(2025b\)](#) provides a canonical specification satisfying all the hypotheses of [Theorem 2](#). It is defined by

$$I_{AD}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v),$$

Table 1: Characterization and axiomatic compliance of a monotone class of depth–span average indices

Family characterization		Description
Functional form		$I(h) = \frac{1}{ V(h) } \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$
Properties of g		The local evaluation function g depends on a node’s depth (s) and span (σ). For the compliance results below, it is weakly increasing in both coordinates and uniformly strictly increasing in at least one coordinate. DE requires the additional representation-level condition indicated below.
Axiomatic Compliance	Status	Reason
A (Anonymity)	✓	Evaluation depends only on structural depth and span, which are invariant under relabeling.
IRP (Intensive Replication)	✓	Replication reproduces the same multiset of depth–span pairs in every copy and leaves the population average unchanged: $I(h^k) = I(h)$.
ERP (Extensive Replication)	✗	Replication leaves the index unchanged rather than increasing it.
CRP (Concentration Replication)	✗	Replication leaves the index unchanged rather than decreasing it.
SR (Subordination Removal)	✓	Subordination removal leaves the population denominator unchanged. Coordinatewise monotonicity prevents local scores from increasing, while uniform strictness makes the decrease strict when the removed edge is non-redundant.
UE* / DE* (Conditional Extensions)	✓	Cross-edges increase at least one depth and at least one span while decreasing no coordinate. Uniform strictness increases the aggregate local score, and the dominance premise prevents population dilution.
DE (Downward Extension)	✓ [†]	Satisfied if the index admits a weakly increasing depth–span representation $\hat{g}$ such that $\hat{g}(D(h) + 1, 0) > I(h)$ for every hierarchy h .
UE (Upward Extension)	✗	Every index in the stated subclass satisfies SR , which is incompatible with UE by Theorem 1 .
[†] Under the sufficient representation-level condition (DB) in Theorem 2 .		

and therefore admits the depth–span representation

$$g_{\text{AD}}(s, \sigma) = s.$$

This function is weakly increasing in both coordinates and uniformly strictly increasing in depth, since

$$g_{\text{AD}}(s + 1, \sigma) - g_{\text{AD}}(s, \sigma) = 1 > 0,$$

while

$$g_{\text{AD}}(s, \sigma + 1) - g_{\text{AD}}(s, \sigma) = 0.$$

It also satisfies (DB). Indeed, for every hierarchy $h \in \mathcal{H}$, writing

$$D(h) = \max_{v \in V(h)} s_h(v),$$

we have

$$I_{\text{AD}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v) \leq D(h) < D(h) + 1 = g_{\text{AD}}(D(h) + 1, 0).$$

Consequently, **Theorem 2** implies that the average depth index satisfies **SR**, **UE***, **DE***, and **DE**. By **Theorem 1**, however, it cannot also satisfy **UE**. Thus, average depth provides an established example for which the sufficient conditions in **Theorem 2** are jointly satisfied, while **UE** fails.

4.2 Depth–span concentration indices

Let

$$\mathcal{H}_{\geq 2} = \{h \in \mathcal{H} : |V(h)| \geq 2\}.$$

We now turn to the concentration normalization. Rather than evaluating hierarchy per person, this approach evaluates hierarchical structure relative to the number of potential pairs of individuals. A population of $|V(h)| \geq 2$ individuals admits exactly

$$\binom{|V(h)|}{2} = \frac{|V(h)|(|V(h)| - 1)}{2}$$

unordered pairs. Since acyclicity permits at most one direction of hierarchical comparability between any two distinct individuals, this pair count provides a finite-population benchmark for ancestor–descendant relations. Dividing the aggregate local score by this number normalizes hierarchy by the number of potential pairs.

Definition 3. A function $I : \mathcal{H}_{\geq 2} \rightarrow \mathbb{R}$ is a *depth–span concentration index* if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \rightarrow \mathbb{R}$ such that, for every $h \in \mathcal{H}_{\geq 2}$,

$$I(h) = \frac{1}{\binom{|V(h)|}{2}} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

Such a function g is called a *depth–span concentration representation* of I .

Every depth–span concentration index satisfies **A** on its domain because relabeling preserves every depth–span pair.

For every $h \in \mathcal{H}_{\geq 2}$ and every integer $k \geq 1$, replication preserves the depth–span pair of every node in every copy. Consequently, writing $N = |V(h)|$,

$$I(h^k) = \frac{1}{\binom{kN}{2}} \sum_{v \in V(h^k)} g(s_{h^k}(v), \sigma_{h^k}(v)) = \frac{k \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))}{\frac{kN(kN-1)}{2}} = \frac{N-1}{kN-1} I(h).$$

Since $\frac{N-1}{kN-1} < 1$ for every $k > 1$ and $N \geq 2$, a depth–span concentration index satisfies **CRP** on its domain if and only if $I(h) > 0$ for every non-flat hierarchy h .

A convenient set of primitive conditions guaranteeing this positivity is $g(0, 0) = 0$, weak coordinatewise monotonicity,

$$g(s+1, \sigma) \geq g(s, \sigma) \quad \text{and} \quad g(s, \sigma+1) \geq g(s, \sigma)$$

for every $s, \sigma \in \mathbb{Z}_+$, and uniform strictness in at least one coordinate. Weak coordinatewise monotonicity gives $g(s, \sigma) \geq g(0, 0) = 0$ for every $s, \sigma \in \mathbb{Z}_+$.

If h is non-flat, it contains an edge $j \rightarrow i$, so

$$\sigma_h(j) \geq 1 \quad \text{and} \quad s_h(i) \geq 1.$$

If uniform strictness holds in depth, then

$$g(s_h(i), \sigma_h(i)) > g(0, \sigma_h(i)) \geq g(0, 0) = 0.$$

If uniform strictness holds in span, then

$$g(s_h(j), \sigma_h(j)) > g(s_h(j), 0) \geq g(0, 0) = 0.$$

Under either condition, all local scores are nonnegative and at least one is strictly positive. Hence, $I(h) > 0$ for every non-flat hierarchy. Therefore, for every non-flat $h \in \mathcal{H}_{\geq 2}$ and every integer $k > 1$,

$$I(h^k) = \frac{N-1}{kN-1} I(h) < I(h),$$

so the index satisfies **CRP** on its domain.

Remark 6 (pairwise and polynomial normalizations). The pairwise normalization

$$\binom{|V(h)|}{2} = \frac{|V(h)|(|V(h)|-1)}{2}$$

differs from squared-population normalization by a constant factor and a lower-order population term. Both normalizations grow quadratically with population size, but they have different finite-population interpretations. The pairwise denominator equals the number of unordered pairs of distinct individuals, whereas $|V(h)|^2$ is a homogeneous quadratic population normalization.

This distinction affects replication formulas. Under the pairwise normalization used here, if $N = |V(h)|$, then replication gives

$$I(h^k) = \frac{N-1}{kN-1}I(h).$$

Thus, a depth–span concentration index that is positive on every non-flat hierarchy satisfies **CRP**. By contrast, normalizing the aggregate local score by $|V(h)|^2$ yields the inverse replication formula $I(h^k) = I(h)/k$. Pairwise normalization therefore preserves the strict concentration replication principle while replacing inverse scaling with the finite-population factor $(N-1)/(kN-1)$. For fixed k , this factor converges to $1/k$ as $N \rightarrow \infty$, so the two normalizations have asymptotically equivalent replication behavior.

Other normalizations may be appropriate if potential pairs are not the relevant structural benchmark. For example, a numerator evaluating higher-order structures, such as directed triads, may call for a cubic population benchmark. Intermediate growth rates may be useful when the relevant relational structure grows faster than linearly but slower than quadratically with population. Such cases arise in empirical network settings in which the number of observed relationships increases nonlinearly with population size (Leskovec et al., 2005). The appropriate benchmark may also depend on cognitive or organizational constraints on the number of relationships that individuals can maintain (Dunbar, 1992; Williamson, 1967).

The density penalty creates a fundamental incompatibility with the extension axioms. Under average normalization, adjoining an identical disconnected hierarchy preserves the index. Under suitable monotonicity and strictness conditions, adding a cross-edge then raises the overall average. Under pairwise normalization, duplicating a large population doubles the unnormalized numerator but increases the pairwise denominator by a factor approaching four. The resulting fragmentation lowers the index whenever the original hierarchy has a positive value. The following results show that this pairwise dilution is incompatible with both the conditional and unrestricted extension principles.

The next result shows that, whenever a depth–span concentration index takes a positive value on some hierarchy, pairwise dilution overpowers the structural gain generated by a single cross-edge between two sufficiently large identical structures. The impossibility does not depend on structural monotonicity or sign restrictions on local evaluations.

Theorem 3. *Let I be a depth–span concentration index with representation g . Suppose there exists a hierarchy $C \in \mathcal{H}$ with $|V(C)| \geq 2$ such that $I(C) > 0$. Then I satisfies neither **UE*** nor **DE***.*

Proof. For every hierarchy $\tilde{h}$ in the domain of I , define

$$S(\tilde{h}) = \sum_{v \in V(\tilde{h})} g(s_{\tilde{h}}(v), \sigma_{\tilde{h}}(v)).$$

Because $I(C) > 0$, we have $S(C) > 0$. Define the constant cross-edge gain

$$\Delta = g(1, 0) + g(0, 1) - 2g(0, 0).$$

For each integer $k \geq 1$, let C^k be the k -fold replication of C . Let $r_k \notin V(C^k)$ and define

$$h_k = C^k \cup \{r_k\},$$

where r_k is isolated. Write $n_k = |V(h_k)|$, and let S_k denote the unnormalized sum of local scores in h_k . Because depth and span are evaluated independently within each disconnected component,

$$S_k = kS(C) + g(0, 0).$$

Since $S(C) > 0$, we have $S_k \rightarrow \infty$ as $k \rightarrow \infty$, for every finite real value of $g(0, 0)$.

Let $h_{o,k}$ be an isomorphic, node-disjoint copy of h_k . Because every depth–span concentration index is anonymous, $I(h_{o,k}) = I(h_k)$. Thus, the weak dominance premise in **UE*** holds with equality.

Form an upward partial attachment $h_k^* \in h_{o,k} \uplus h_k$ by adding a single edge from the isolated node of $h_{o,k}$ to the isolated node r_k of h_k . The copies of C remain unchanged, and no cross-edge is incident to any of their nodes.

Before the attachment, the two isolated nodes each contribute $g(0, 0)$. After the attachment, they form a two-node directed chain. The upper node has depth–span pair $(0, 1)$, while the lower node has pair $(1, 0)$. Their combined contribution therefore changes by exactly

$$\Delta = g(0, 1) + g(1, 0) - 2g(0, 0).$$

Hence,

$$S(h_k^*) = 2S_k + \Delta.$$

Since $|V(h_k^*)| = 2n_k$, the concentration index of the attached hierarchy is

$$I(h_k^*) = \frac{2S_k + \Delta}{\binom{2n_k}{2}} = \frac{2S_k + \Delta}{n_k(2n_k - 1)}.$$

The baseline index is

$$I(h_k) = \frac{S_k}{\binom{n_k}{2}} = \frac{2S_k}{n_k(n_k - 1)}.$$

The inequality

$$I(h_k^*) < I(h_k)$$

is therefore equivalent to

$$\frac{2S_k + \Delta}{n_k(2n_k - 1)} < \frac{2S_k}{n_k(n_k - 1)}.$$

Multiplying both sides by $n_k(n_k - 1)(2n_k - 1)$ gives

$$(2S_k + \Delta)(n_k - 1) < 2S_k(2n_k - 1).$$

Expanding and rearranging terms yields

$$\Delta(n_k - 1) < 2S_k n_k,$$

or equivalently,

$$\Delta \left(1 - \frac{1}{n_k}\right) < 2S_k.$$

The left-hand side is bounded above by $\max\{\Delta, 0\}$, while $2S_k \rightarrow \infty$. Hence, the strict inequality holds for every sufficiently large k . Consequently, $I(h_k^*) < I(h_k)$, which contradicts **UE***.

For **DE***, reverse the direction of the single cross-edge, so that it points from r_k in the upper base h_k to the isolated node in the lower hierarchy $h_{o,k}$. This gives an admissible attachment $h_k^* \in h_k \uplus h_{o,k}$. The dominance premise again holds with equality: $I(h_{o,k}) = I(h_k)$. Reversing the edge interchanges the depth–span pairs $(0, 1)$ and $(1, 0)$ of the two isolated nodes, so the unnormalized sum remains

$$S(h_k^*) = 2S_k + \Delta.$$

The same inequality therefore gives $I(h_k^*) < I(h_k)$, contradicting **DE***. $\blacksquare$

Having established the failure of the conditional extensions, we now show that depth–span concentration indices are also incompatible with the unrestricted vertical extensions **UE** and **DE**.

Because concentration indices are defined on $\mathcal{H}_{\geq 2}$, their compliance with an extension axiom is understood relative to this domain: the axiom applies whenever every index value appearing in its statement is defined.

Under concentration normalization, population growth imposes a pairwise dilution penalty. Because the unrestricted extension axioms quantify over every admissible base and attachment, they force bounds on average unnormalized scores in the hierarchies used in the dilution constructions. Pairwise normalization then drives the normalized values of expanding hierarchies to zero, contradicting any fixed positive baseline. No monotonicity, sign, or growth restriction on the local evaluation function is required.

For every $h \in \mathcal{H}$, write

$$D(h) = \max_{v \in V(h)} s_h(v)$$

for the maximum depth of h .

Theorem 4. *Let I be a depth–span concentration index. Suppose there exists a hierarchy $C \in \mathcal{H}_{\geq 2}$ such that $I(C) > 0$. Then I satisfies neither **UE** nor **DE** on its domain.*

Proof. Fix a depth–span representation g of I , and define

$$S(h) = \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$$

for every $h \in \mathcal{H}$. Thus, whenever $h \in \mathcal{H}_{\geq 2}$,

$$I(h) = \frac{S(h)}{\binom{|V(h)|}{2}} = \frac{2S(h)}{|V(h)|(|V(h)| - 1)}.$$

We first prove the failure of **UE**. Suppose, toward a contradiction, that I satisfies **UE** on its domain.

Step 1: a universal upper bound. Fix an arbitrary hierarchy $H \in \mathcal{H}$, and write

$$n = |V(H)| \quad \text{and} \quad S = S(H)$$

for its size and unnormalized local-score sum. For each integer $k \geq 1$, let H^k be the k -fold replication of H . Let $r_k \notin V(H^k)$ and define $h_k = H^k \cup \{r_k\}$, where r_k is isolated. Let $u \notin V(h_k)$ and set $h_o = \{u\}$. Form $h_k^* \in h_o \oplus h_k$ by adding the single edge $u \rightarrow r_k$. This is an admissible upward attachment because u is the unique leaf of h_o and r_k is a root of h_k .

The k copies of H remain unchanged. Hence, $|V(h_k)| = kn + 1$, and

$$I(h_k) = \frac{2[kS + g(0, 0)]}{(kn + 1)kn}.$$

Since $|V(h_k^*)| = kn + 2$, we have

$$I(h_k^*) = \frac{2[kS + g(0, 1) + g(1, 0)]}{(kn + 2)(kn + 1)}.$$

By **UE**, $I(h_k^*) > I(h_k)$ for every $k \geq 1$. Multiplying both sides by the positive quantity $(kn + 1)/2$ gives

$$\frac{kS + g(0, 1) + g(1, 0)}{kn + 2} > \frac{kS + g(0, 0)}{kn}.$$

Cross-multiplication yields

$$kn[kS + g(0, 1) + g(1, 0)] > (kn + 2)[kS + g(0, 0)].$$

Expanding both sides produces

$$k^2nS + kn[g(0, 1) + g(1, 0)] > k^2nS + 2kS + (kn + 2)g(0, 0).$$

Subtracting k^2nS and dividing by k gives

$$n[g(0, 1) + g(1, 0)] > 2S + \left(n + \frac{2}{k}\right)g(0, 0).$$

Letting $k \rightarrow \infty$ yields

$$n[g(0, 1) + g(1, 0)] \geq 2S + ng(0, 0).$$

Consequently,

$$\frac{S(H)}{|V(H)|} \leq \frac{g(0, 1) + g(1, 0) - g(0, 0)}{2}.$$

Define

$$\Delta_{\text{UE}} = \frac{g(0, 1) + g(1, 0) - g(0, 0)}{2}.$$

Since H was arbitrary,

$$\frac{S(H)}{|V(H)|} \leq \Delta_{\text{UE}} \tag{8}$$

for every $H \in \mathcal{H}$.

Step 2: contradiction. For each integer $m \geq 1$, let h_o^m be a flat hierarchy of m isolated nodes that is node-disjoint from C . Fix a root j of C , and form $H_m \in h_o^m \oplus C$ by adding an edge from every node of h_o^m to j .

By **UE**,

$$I(H_m) > I(C) > 0.$$

On the other hand, applying (8) to H_m gives

$$I(H_m) = \frac{2}{|V(H_m)| - 1} \frac{S(H_m)}{|V(H_m)|} \leq \frac{2\Delta_{\text{UE}}}{|V(C)| + m - 1}.$$

Applying (8) to C shows that

$$\Delta_{\text{UE}} \geq \frac{S(C)}{|V(C)|} = \frac{|V(C)| - 1}{2} I(C) > 0.$$

Choose m sufficiently large that

$$\frac{2\Delta_{\text{UE}}}{|V(C)| + m - 1} < I(C).$$

Then $I(H_m) < I(C)$, contradicting **UE**. Therefore, I does not satisfy **UE** on its domain.

We next prove the failure of **DE**. Suppose, toward a contradiction, that I satisfies **DE** on its domain.

For each integer $m \geq 1$, let h_o^m be a flat hierarchy of m isolated nodes that is node-disjoint from C . Fix a maximum-depth leaf ℓ of C , and form $H_m \in C \boxplus h_o^m$ by adding an edge from ℓ to every node of h_o^m .

Write $D = D(C) + 1$. Every appended node has depth D , while every original node of C has depth at most $D(C)$. Therefore, $D(H_m) = D$ for every $m \geq 1$. By **DE**,

$$I(H_m) > I(C) > 0.$$

Let B be a directed chain with D edges and hence $D + 1$ nodes. Its unique leaf has depth exactly D . Let $w \notin V(B)$, and let B^+ be obtained by adding an edge from the unique leaf of B to w . Define

$$b = |V(B)|, \quad T = S(B), \quad \delta_B = S(B^+) - S(B).$$

Because $D = D(C) + 1$ does not depend on m , neither B nor δ_B depends on m .

Fix $m \geq 1$. For each integer $k \geq 1$, take copies of H_m that are node-disjoint from B^+ and form $h_{m,k} = (H_m)^k \cup B$. Both $(H_m)^k$ and B have maximum depth D . Hence, the unique leaf of B is a maximum-depth leaf of $h_{m,k}$. Form $h_{m,k}^* \in h_{m,k} \boxplus \{w\}$ by adding the edge from the unique leaf of B to w .

Write

$$n_m = |V(H_m)| \quad \text{and} \quad S_m = S(H_m).$$

The attachment changes only the subgraph on $V(B) \cup \{w\}$. Since $|V(h_{m,k})| = kn_m + b$, we have

$$I(h_{m,k}) = \frac{2(kS_m + T)}{(kn_m + b)(kn_m + b - 1)}.$$

Since $|V(h_{m,k}^*)| = kn_m + b + 1$, we have

$$I(h_{m,k}^*) = \frac{2(kS_m + T + \delta_B)}{(kn_m + b + 1)(kn_m + b)}.$$

By **DE**, $I(h_{m,k}^*) > I(h_{m,k})$ for every $k \geq 1$. Multiplying by $(kn_m + b)/2$ gives

$$\frac{kS_m + T + \delta_B}{kn_m + b + 1} > \frac{kS_m + T}{kn_m + b - 1}.$$

Cross-multiplication yields

$$(kS_m + T + \delta_B)(kn_m + b - 1) > (kS_m + T)(kn_m + b + 1).$$

Expanding the left side and subtracting $(kS_m + T)(kn_m + b - 1)$ gives

$$\delta_B(kn_m + b - 1) > (kS_m + T)[(kn_m + b + 1) - (kn_m + b - 1)] = 2(kS_m + T).$$

Dividing by k yields

$$\delta_B \left(n_m + \frac{b-1}{k} \right) > 2 \left(S_m + \frac{T}{k} \right).$$

Letting $k \rightarrow \infty$ yields $n_m \delta_B \geq 2S_m$. Hence, $\frac{S_m}{n_m} \leq \frac{\delta_B}{2}$.

Because

$$I(H_m) > I(C) > 0,$$

we have $S_m > 0$. The preceding inequality therefore implies $\delta_B > 0$. Consequently,

$$I(H_m) = \frac{2}{n_m - 1} \frac{S_m}{n_m} \leq \frac{\delta_B}{n_m - 1} = \frac{\delta_B}{|V(C)| + m - 1}.$$

Choose m sufficiently large that

$$\frac{\delta_B}{|V(C)| + m - 1} < I(C).$$

Then $I(H_m) < I(C)$, contradicting **DE**. Therefore, I does not satisfy **DE** on its domain. ■

Subordination removal. Pairwise normalization does not affect the subordination-removal argument: the node set is unchanged, so the denominator is unchanged. The response of the index is therefore determined entirely by the change in the unnormalized sum of local scores.

Proposition 1. *Let I be a depth-span concentration index with representation*

$$I(h) = \frac{1}{\binom{|V(h)|}{2}} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)), \quad |V(h)| \geq 2.$$

*Suppose that g is weakly increasing in both coordinates and that at least one of **(SD)** and **(SS)** holds. Then I satisfies **SR** on its domain.*

Proof. Subordination removal creates no new ancestor-descendant pair because each bypass edge replaces a path already present in the original hierarchy. Hence, if h' is obtained from h by removing a subordination relation, then, for every node v ,

$$s_{h'}(v) \leq s_h(v) \quad \text{and} \quad \sigma_{h'}(v) \leq \sigma_h(v).$$

Weak coordinatewise monotonicity implies that no local score increases. Since $V(h') = V(h)$, the pairwise denominator is unchanged, and therefore $I(h') \leq I(h)$.

If the removed edge (j, i) is non-redundant, then j no longer reaches i after the removal. Thus,

$$s_{h'}(i) < s_h(i) \quad \text{and} \quad \sigma_{h'}(j) < \sigma_h(j).$$

If **(SD)** holds, the strict decrease in the depth of i , together with the weak decrease in its span, strictly lowers its local score. If **(SS)** holds, the strict decrease in the span of j , together with the weak decrease in its depth, strictly lowers its local score. In either case, at least one local score decreases strictly while none increases. Hence, $I(h') < I(h)$. Thus, I satisfies **SR** on its domain. ■

The normalization and monotonicity conditions discussed above define a nontrivial class of depth–span concentration indices satisfying **A**, **SR**, and **CRP** on their domain. The normalization $g(0, 0) = 0$ helps guarantee that the index assigns a strictly positive value to every non-flat hierarchy and hence satisfies the strict inequality required by **CRP**. It is not needed for **SR**.

The extension axioms exhibit the opposite pattern. Since the preceding conditions assign a strictly positive index value to every non-flat hierarchy, **Theorem 3** and **Theorem 4** apply. Consequently, no index in this class satisfies **UE***, **DE***, **UE**, or **DE** on its domain. Thus, the class defined by the preceding normalization, monotonicity, and strictness conditions satisfies **A**, **SR**, and **CRP**, but no extension axiom considered here.

Remark 7 (admissible specifications and equivalent representations). The conditions above permit sublinear, linear, and superlinear local evaluations. For example, the functions

$$\begin{aligned} g_1(s, \sigma) &= \sqrt{s} + \sqrt{\sigma}, \\ g_2(s, \sigma) &= as + b\sigma, \quad a, b > 0, \\ g_3(s, \sigma) &= s^2 + \sigma^2 + s + \sigma \end{aligned}$$

all satisfy $g(0, 0) = 0$, are weakly increasing in both coordinates, and are uniformly strictly increasing in both coordinates. Hence, each defines a depth–span concentration index satisfying **SR** and **CRP** on its domain. By **Theorem 3** and **Theorem 4**, none satisfies **UE***, **DE***, **UE**, or **DE** on its domain.

Equivalent-representation issues are the same as in the average case, except that additive constants are not innocuous under pairwise normalization. If g represents a concentration index I , then, for every $\beta > 0$ and $\theta \in \mathbb{R}$, the function

$$\tilde{g}(s, \sigma) = \beta g(s, \sigma) + \theta(\sigma - s)$$

represents the positive scalar transformation $\tilde{I}(h) = \beta I(h)$. Indeed, every hierarchy satisfies

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

so the zero-sum term $\theta(\sigma - s)$ does not affect the aggregate local score. In particular, when $\beta = 1$, $\tilde{g}$ represents the same concentration index as g .

By contrast, for every $h \in \mathcal{H}_{\geq 2}$, adding a constant γ to every local score changes the index by

$$\frac{1}{\binom{|V(h)|}{2}} \sum_{v \in V(h)} (g(s_h(v), \sigma_h(v)) + \gamma) = I(h) + \frac{\gamma |V(h)|}{\binom{|V(h)|}{2}} = I(h) + \frac{2\gamma}{|V(h)| - 1}.$$

Such a shift can alter rankings across hierarchies of different sizes, change replication behavior, and destroy the normalization $g(0, 0) = 0$ unless $\gamma = 0$.

Table 2 summarizes these conclusions.

4.3 Depth–span total indices

We next consider total aggregation, which evaluates aggregate hierarchical structure by summing individual depth–span scores across the organization. Unlike the average and concentration normalizations, this approach makes no adjustment for population size.

Table 2: Characterization and axiomatic compliance of a monotone class of depth–span concentration indices on $\mathcal{H}_{\geq 2}$

Family characterization		Description
Functional form		$I(h) = \frac{1}{\binom{ V(h) }{2}} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)), \text{ for } h \in \mathcal{H}_{\geq 2}$
Properties of g		For the compliance results below, the local evaluation function g satisfies $g(0, 0) = 0$, is weakly increasing in both coordinates, and is uniformly strictly increasing in at least one coordinate.
Axiomatic Compliance	Status	Reason
A (Anonymity)	✓	Evaluation depends only on structural depth and span, which are invariant under relabeling.
IRP (Intensive Replication)	✗	For every non-flat hierarchy with $N = V(h) $ and every $k > 1$, $I(h^k) = \frac{N-1}{kN-1} I(h) < I(h)$, rather than $I(h^k) = I(h)$.
ERP (Extensive Replication)	✗	Replication decreases rather than increases the value of every non-flat hierarchy.
CRP (Concentration Replication)	✓	Replication gives $I(h^k) = \frac{N-1}{kN-1} I(h)$, which is strictly smaller than $I(h)$ for every $k > 1$. The zero baseline, coordinatewise monotonicity, and uniform strictness ensure that $I(h) > 0$ for every non-flat hierarchy.
SR (Subordination Removal)	✓	Subordination removal leaves the node set, and hence the pairwise denominator, unchanged. Coordinatewise monotonicity prevents local scores from increasing, while uniform strictness makes the decrease strict when the removed edge is non-redundant.
UE* / DE* (Conditional Extensions)	✗	Pairwise normalization turns the duplication of a positive hierarchy into dilution that cannot be offset by a single sparse cross-edge.
DE (Downward Extension)	✗	Background replication forces a fixed-depth average-score ceiling, contradicting the strict increase required along expanding downward attachments of a positive base.
UE (Upward Extension)	✗	The axiom would force a universal ceiling on average unnormalized scores, contradicting the strict increase required along expanding upward attachments of a positive base.

Definition 4. A hierarchy index $I : \mathcal{H} \rightarrow \mathbb{R}$ is a *depth–span total index* if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \rightarrow \mathbb{R}$ such that, for every $h \in \mathcal{H}$,

$$I(h) = \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)).$$

Such a function g is called a *depth–span total representation* of I .

Every depth–span total index satisfies **A** because relabeling preserves every depth–span pair. It also obeys a linear replication rule. For every $h \in \mathcal{H}$ and every integer $k \geq 1$, replication preserves the depth–span pair of every node in every copy. Consequently,

$$I(h^k) = \sum_{v \in V(h^k)} g(s_{h^k}(v), \sigma_{h^k}(v)) = k \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)) = kI(h).$$

Thus, a depth–span total index satisfies **ERP** if and only if $I(h) > 0$ for every non-flat hierarchy h .

Theorem 5. Let I be a depth–span total index with representation g . Suppose that g is weakly increasing in both coordinates, satisfies $g(0, 0) = 0$, and satisfies at least one of **(SD)** and **(SS)**. Then I satisfies **ERP**, **SR**, **UE**, **DE**, **UE***, and **DE***.

Proof. Weak coordinatewise monotonicity and $g(0, 0) = 0$ imply $g(s, \sigma) \geq 0$ for every $s, \sigma \in \mathbb{Z}_+$. Hence, every hierarchy has a nonnegative total score.

We first prove positivity on non-flat hierarchies. Let h be non-flat, and choose an edge $j \rightarrow i$. Then

$$\sigma_h(j) \geq 1 \quad \text{and} \quad s_h(i) \geq 1.$$

If **(SD)** holds, repeated strictness in depth gives

$$g(s_h(i), \sigma_h(i)) > g(0, \sigma_h(i)) \geq 0.$$

If **(SS)** holds, repeated strictness in span gives

$$g(s_h(j), \sigma_h(j)) > g(s_h(j), 0) \geq 0.$$

Thus, all local scores are nonnegative and at least one is strictly positive, so $I(h) > 0$. Together with the replication identity $I(h^k) = kI(h)$, this proves **ERP**.

We next prove **SR**. Let h' be obtained from h by removing a subordination relation. As in **Theorem 2**, the operation creates no new ancestor–descendant pair. Hence, for every node v ,

$$s_{h'}(v) \leq s_h(v) \quad \text{and} \quad \sigma_{h'}(v) \leq \sigma_h(v).$$

Weak coordinatewise monotonicity gives $I(h') \leq I(h)$. If the removed edge (j, i) is non-redundant, then j no longer reaches i after the removal, so

$$s_{h'}(i) < s_h(i) \quad \text{and} \quad \sigma_{h'}(j) < \sigma_h(j).$$

If **(SD)** holds, the strict decrease in the depth of i , together with the weak decrease in its span, strictly lowers its local score. If **(SS)** holds, the strict decrease in the span of j , together with the weak

decrease in its depth, strictly lowers its local score. In either case, at least one local score decreases strictly while none increases. Hence, $I(h') < I(h)$, proving **SR**.

It remains to prove the extension axioms. Consider any partial attachment $h^* \in h_1 \uplus h_2$. The attachment adds at least one cross-edge and removes no ancestor–descendant relation. Thus, no node’s depth or span decreases. Every cross-edge increases the span of its source and the depth of its target. Under either **(SD)** or **(SS)**, at least one local score increases strictly while none decreases. Therefore,

$$I(h^*) > I(h_1) + I(h_2).$$

Since every hierarchy has a nonnegative total score, it follows that

$$I(h^*) > I(h_2) \quad \text{and} \quad I(h^*) > I(h_1).$$

Applying this observation to $h_o \uplus h$ proves **UE***, and applying it to $h \uplus h_o$ proves **DE***. Since

$$h_o \oplus h \subseteq h_o \uplus h \quad \text{and} \quad h \boxplus h_o \subseteq h \uplus h_o,$$

the same argument proves **UE** and **DE**. ■

The normalization $g(0, 0) = 0$, together with weak coordinatewise monotonicity, ensures that every hierarchy has a nonnegative total score. Total aggregation therefore avoids the dilution effects that drive the average and concentration impossibility results. Under the stated assumptions, every admissible partial attachment raises the total index strictly above the index of each input hierarchy, so the dominance premises in **UE*** and **DE*** are not needed.

Remark 8 (representations). Representation issues parallel those for the average and concentration families. If g represents a total index I , then, for every $\beta > 0$ and $\theta \in \mathbb{R}$, the function

$$\tilde{g}(s, \sigma) = \beta g(s, \sigma) + \theta(\sigma - s)$$

represents the positive scalar transformation $\tilde{I}(h) = \beta I(h)$. Indeed, every hierarchy satisfies

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

so the zero-sum term $\theta(\sigma - s)$ does not change the aggregate score. In particular, when $\beta = 1$, $\tilde{g}$ is an alternative representation of the same total index.

Adding a constant to every local score does not preserve the index. If

$$\tilde{g}(s, \sigma) = g(s, \sigma) + \gamma,$$

then

$$\tilde{I}(h) = I(h) + \gamma|V(h)|.$$

The change depends on population size and may alter rankings across hierarchies of different sizes. It also violates $\tilde{g}(0, 0) = 0$ unless $\gamma = 0$.

The preceding theorem identifies a nontrivial class of depth–span total indices satisfying **A**, **ERP**, **SR**, **UE**, **DE**, **UE***, and **DE***. **Table 3** summarizes these conclusions.

Table 3: Characterization and axiomatic compliance of a monotone class of depth–span total indices

Family characterization		Description
Functional form		$I(h) = \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$
Properties of g		For the compliance results below, the local evaluation function g satisfies $g(0, 0) = 0$, is weakly increasing in both coordinates, and is uniformly strictly increasing in at least one coordinate.
Axiomatic Compliance	Status	Reason
A (Anonymity)	✓	Evaluation depends only on structural depth and span, which are invariant under relabeling.
IRP (Intensive Replication)	✗	For every non-flat hierarchy and every $k > 1$, $I(h^k) = kI(h) > I(h)$, rather than $I(h^k) = I(h)$.
ERP (Extensive Replication)	✓	Replication gives $I(h^k) = kI(h)$. The zero baseline, coordinatewise monotonicity, and uniform strictness ensure that $I(h) > 0$ for every non-flat hierarchy.
CRP (Concentration Replication)	✗	Replication increases rather than decreases the value of every non-flat hierarchy.
SR (Subordination Removal)	✓	Coordinatewise monotonicity prevents local scores from increasing after subordination removal, while uniform strictness makes the decrease strict when the removed edge is non-redundant.
UE* / DE* (Conditional Extensions)	✓	Every partial attachment creates at least one strict local-score gain and no loss. Both input hierarchies have nonnegative total scores, so the attached hierarchy strictly exceeds each input hierarchy without using the dominance premise.
DE (Downward Extension)	✓	A downward attachment strictly raises at least one local score and lowers none. The lower hierarchy has a nonnegative total score, so the attached hierarchy strictly exceeds the base.
UE (Upward Extension)	✓	An upward attachment strictly raises at least one local score and lowers none. The upper hierarchy has a nonnegative total score, so the attached hierarchy strictly exceeds the base.

5 Hierarchy indices from the literature

This section evaluates several established hierarchy indices using the axioms introduced above. The purpose is not to rederive these measures but to compare their responses to relabeling, replication, subordination removal, and vertical extension.

Several indices considered below are not defined on every hierarchy in $\mathcal{H}$. For each such index and each axiom, compliance is evaluated relative to the index's domain: the axiom is applied whenever all hierarchies whose index values appear in its statement belong to that domain. Any satisfaction or violation statement is understood in this restricted sense.

Detailed calculations and representative counterexamples are presented first. Table 4 summarizes the results at the end of the section.

5.1 Average depth

The average depth index studied by Carbonell-Nicolau (2025b) is

$$I_s(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v).$$

Thus, $I_s(h)$ is the expected number of direct and indirect supervisors of a uniformly selected individual.

The index is a depth–span average index with representation $g(s, \sigma) = s$. This representation is weakly increasing in both coordinates and uniformly strictly increasing in depth:

$$g(s + 1, \sigma) - g(s, \sigma) = 1 > 0.$$

Moreover,

$$g(s, \sigma + 1) - g(s, \sigma) = 0.$$

The index satisfies **A** because relabeling preserves every node's depth. It also satisfies **IRP** because replication reproduces the same multiset of depths in every copy: $I_s(h^k) = I_s(h)$ for every $h \in \mathcal{H}$ and every integer $k \geq 1$. Consequently, it violates **ERP** and **CRP** on non-flat hierarchies.

By Theorem 2, weak coordinatewise monotonicity and uniform strictness in depth imply that I_s satisfies **SR**, **UE***, and **DE***.

The average depth index also satisfies the sufficient representation-level condition (**DB**). Indeed, for every hierarchy $h \in \mathcal{H}$,

$$I_s(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v) \leq D(h),$$

where

$$D(h) = \max_{v \in V(h)} s_h(v).$$

Consequently,

$$g(D(h) + 1, 0) = D(h) + 1 > I_s(h).$$

Therefore, I_s satisfies **DE**.

Because I_s satisfies **SR**, **Theorem 1** implies that it violates **UE**. Thus, average depth satisfies **A**, **IRP**, **SR**, **UE***, **DE***, and **DE**. It violates **ERP**, **CRP**, and **UE**.

The identity

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v)$$

provides an alternative interpretation. Every ordered ancestor–descendant pair contributes once to the depth of its lower node and once to the span of its upper node. Hence,

$$I_s(h) = \frac{1}{2|V(h)|} \sum_{v \in V(h)} (s_h(v) + \sigma_h(v)).$$

Average depth is therefore one half of the average combined upward and downward reach.

5.2 Global reaching centrality

The global reaching centrality index of **Mones et al. (2012)** is

$$\text{GRC}(h) = \frac{1}{N-1} \sum_{v \in V(h)} (C_R^{\max} - C_R(v)),$$

where

$$N = |V(h)|, \quad C_R(v) = \frac{|\text{Desc}_h(v)|}{N-1}, \quad \text{and} \quad C_R^{\max} = \max_{u \in V(h)} C_R(u).$$

The formula requires $N \geq 2$. GRC measures the dispersion of downward reach relative to the node with maximum reach.

The GRC index satisfies **A** because relabeling preserves every descendant count and therefore the multiset of reaching centralities.

Replication. Let h have $N \geq 2$ nodes. In the k -fold replication h^k , each node (v, ℓ) has the same descendant count as v has in h , while the population denominator changes from $N-1$ to $kN-1$. Thus, for every $v \in V(h)$ and $\ell \in \{1, \dots, k\}$,

$$C_R^{(h^k)}(v, \ell) = \frac{|\text{Desc}_h(v)|}{kN-1} = \frac{N-1}{kN-1} C_R^{(h)}(v).$$

The maximum reaching centrality scales by the same factor. Consequently,

$$\text{GRC}(h^k) = \frac{k(N-1)^2}{(kN-1)^2} \text{GRC}(h).$$

For every $N \geq 2$ and $k > 1$,

$$0 < \frac{k(N-1)^2}{(kN-1)^2} < 1.$$

Moreover, $\text{GRC}(h) > 0$ for every non-flat hierarchy h . Indeed, every finite DAG has a leaf, whose reaching centrality is zero, while a non-flat hierarchy has a node with at least one descendant and hence positive reaching centrality. Thus, $C_R^{\max} > 0$, and the leaf contributes a strictly positive term to the GRC sum. Replication therefore strictly lowers GRC on every non-flat hierarchy. On its domain, the GRC index satisfies **CRP** and violates **IRP** and **ERP**.

Subordination removal. The GRC index violates **SR**. Let h be the three-node chain

$$1 \rightarrow 2 \rightarrow 3.$$

The reaching centralities are

$$C_R(1) = 1, \quad C_R(2) = \frac{1}{2}, \quad C_R(3) = 0,$$

so

$$\text{GRC}(h) = \frac{1}{2} \left[(1 - 1) + \left(1 - \frac{1}{2}\right) + (1 - 0) \right] = \frac{3}{4}.$$

Remove the non-redundant edge $(2, 3)$. The subordination-removal operation adds the bypass edge $(1, 3)$, so the resulting hierarchy h' has edges

$$1 \rightarrow 2 \quad \text{and} \quad 1 \rightarrow 3.$$

Now,

$$C_R(1) = 1, \quad C_R(2) = C_R(3) = 0,$$

and hence

$$\text{GRC}(h') = \frac{1}{2} [(1 - 1) + (1 - 0) + (1 - 0)] = 1 > \frac{3}{4} = \text{GRC}(h).$$

Thus, removing a non-redundant subordination relation increases GRC, in violation of **SR**.

Unrestricted extensions. Let h again be the three-node chain, so that $\text{GRC}(h) = \frac{3}{4}$. Append a single node ℓ above its root, producing the four-node chain

$$\ell \rightarrow 1 \rightarrow 2 \rightarrow 3.$$

This is an admissible upward attachment. The reaching centralities are $1, \frac{2}{3}, \frac{1}{3}, 0$. Therefore,

$$\text{GRC}(h^*) = \frac{1}{3} \left[(1 - 1) + \left(1 - \frac{2}{3}\right) + \left(1 - \frac{1}{3}\right) + (1 - 0) \right] = \frac{2}{3} < \frac{3}{4}.$$

Hence, the GRC index violates **UE**.

Appending a single node below the deepest leaf instead produces the four-node chain

$$1 \rightarrow 2 \rightarrow 3 \rightarrow \ell.$$

The same calculation gives

$$\text{GRC}(h^*) = \frac{2}{3} < \frac{3}{4}.$$

Thus, the GRC index also violates **DE**.

Conditional extensions. For an N -node directed chain,

$$\text{GRC} = \frac{N}{2(N-1)}.$$

Take two node-disjoint three-node chains h_o and h . Then

$$\text{GRC}(h_o) = \text{GRC}(h) = \frac{3}{4},$$

so the dominance premise in **UE*** holds with equality. Add a single edge from the leaf of h_o to the root of h . The resulting hierarchy is a six-node chain, and therefore

$$\text{GRC}(h^*) = \frac{6}{2(6-1)} = \frac{3}{5} < \frac{3}{4}.$$

Thus, the GRC index violates **UE***.

For **DE***, take one three-node chain as the upper base and the other as the lower appended hierarchy. Add an edge from the leaf of the upper chain to the root of the lower chain. The resulting hierarchy is again a six-node chain. The dominance premise holds with equality, while the same calculation gives

$$\text{GRC}(h^*) = \frac{3}{5} < \frac{3}{4}.$$

Thus, the GRC index violates **DE***.

5.3 Krackhardt's graph-theoretic dimensions

The four graph-theoretic dimensions proposed by **Krackhardt (1994)** measure connectedness, cyclicity, link redundancy, and least-upper-bound structure. On the present domain of finite directed acyclic graphs, they capture features distinct from depth and span.

Ignoring edge directions yields the underlying undirected graph of a hierarchy. Its connected components are the *weakly connected components*, or *weak components*, of the hierarchy. A hierarchy is *weakly connected* if its underlying undirected graph is connected.

5.3.1 Connectedness

For a hierarchy h with $n = |V(h)| \geq 2$, define

$$C(h) = \frac{|\{(i, j) \in V(h) \times V(h) : i \neq j \text{ and } i, j \text{ lie in the same weak component}\}|}{n(n-1)}.$$

Equivalently, if the weakly connected components have sizes $n_1, \dots, n_q$, then

$$C(h) = \frac{\sum_{\ell=1}^q n_{\ell}(n_{\ell}-1)}{n(n-1)}.$$

The expression is undefined at $n = 1$ unless an additional convention is imposed.

The connectedness index satisfies **A** because relabeling preserves weak components and their sizes.

Replication. Let

$$S(h) = \sum_{\ell=1}^q n_{\ell}(n_{\ell} - 1).$$

Replication produces k copies of every weak component, so $S(h^k) = kS(h)$. Consequently,

$$C(h^k) = \frac{kS(h)}{kn(kn - 1)} = \frac{n - 1}{kn - 1}C(h).$$

If h is non-flat, at least one weak component contains two or more nodes, and therefore $C(h) > 0$. For $k > 1$,

$$\frac{n - 1}{kn - 1} < 1.$$

Thus, connectedness satisfies **CRP** and violates **IRP** and **ERP** on its domain.

Subordination removal and extensions. Connectedness violates **SR** because it can remain unchanged after the removal of a non-redundant edge. Let h be the chain

$$1 \rightarrow 2 \rightarrow 3.$$

It is weakly connected, so $C(h) = 1$. Remove the non-redundant edge $(2, 3)$. The bypass rule adds $(1, 3)$, and the resulting hierarchy h' has edges

$$1 \rightarrow 2 \quad \text{and} \quad 1 \rightarrow 3.$$

It remains weakly connected, so $C(h') = 1 = C(h)$, contradicting the strict part of **SR**.

Connectedness also violates **UE**, **DE**, **UE***, and **DE***. To see this, let h_o and h be weakly connected hierarchies. Then $C(h_o) = C(h) = 1$. Every admissible attachment adds at least one cross-edge. Because both input hierarchies are weakly connected, the attached hierarchy is also weakly connected. Hence, $C(h^*) = 1 = C(h)$, contradicting the strict conclusion of each extension axiom. For **UE*** and **DE***, the dominance premise holds with equality.

5.3.2 Graph hierarchy

Krackhardt's graph hierarchy index measures the extent to which reachability is unidirectional. Write

$$H(h) = 1 - \frac{P(h)}{R(h)},$$

where $P(h)$ is the number of unordered pairs of distinct nodes that are mutually reachable and $R(h)$ is the number of unordered pairs for which at least one node reaches the other. The formula is defined when $R(h) > 0$.

On a directed acyclic graph, mutual reachability between distinct nodes is impossible. Indeed, if

$$i \rightsquigarrow j \quad \text{and} \quad j \rightsquigarrow i,$$

the two paths form a directed cycle. Therefore, $P(h) = 0$ for every hierarchy $h \in \mathcal{H}$. Whenever $R(h) > 0$, it follows that $H(h) = 1$.

Thus, graph hierarchy is constant on the non-flat portion of the present domain. It satisfies **A** and **IRP** there trivially because it is constant on its domain. Because it remains constant under replication, it violates **ERP** and **CRP**.

Constancy also implies failure of every strict structural axiom whenever all index values required by the axiom are defined. In particular, the index cannot decrease strictly after the removal of a non-redundant edge or increase strictly after an admissible extension. It therefore violates **SR**, **UE**, **DE**, **UE***, and **DE*** on its domain.

If h is flat, then $R(h) = 0$, so the formula is undefined. In particular, graph hierarchy is not defined on the single-node hierarchy without an additional convention.

5.3.3 Graph efficiency

Krackhardt's graph efficiency index evaluates redundancy in reporting links relative to the minimum required for weak connectivity. Let the weakly connected components of h be indexed by $c = 1, \dots, q$, and let component c contain n_c nodes and m_c directed edges. Under the componentwise formulation used here, define

$$E(h) = 1 - \frac{\sum_{c=1}^q (m_c - (n_c - 1))}{\sum_{c=1}^q \frac{(n_c - 1)(n_c - 2)}{2}},$$

whenever the denominator is positive and the adopted componentwise normalization is well-defined.

The numerator records links in excess of the $n_c - 1$ links needed for weak connectivity within each component. A directed acyclic graph on n_c nodes can have at most $n_c(n_c - 1)/2$ edges. Relative to the $n_c - 1$ edges needed for weak connectivity, its maximum redundancy is therefore $(n_c - 1)(n_c - 2)/2$, as recorded in the denominator. Singleton components contribute zero to both sums and do not affect the formula when the total denominator is positive.

Graph efficiency satisfies **A** because relabeling preserves component sizes and edge counts.

Replication. Replication multiplies both the numerator and denominator in the definition by the same factor. Therefore, $E(h^k) = E(h)$ whenever both sides are defined. Graph efficiency consequently satisfies **IRP** and violates **ERP** and **CRP** on its domain.

Subordination removal and extensions. Graph efficiency violates **SR**. Let h be the three-node chain

$$j \rightarrow i \rightarrow k.$$

Removing the non-redundant edge (i, k) produces the hierarchy h' with edges

$$j \rightarrow i \quad \text{and} \quad j \rightarrow k.$$

Both hierarchies are weakly connected and have three nodes and two edges. Both are trees in their underlying undirected form and therefore have efficiency one under the stated formulation: $E(h') = E(h) = 1$. This violates the strict part of **SR**.

The same construction yields violations of **UE** and **DE**. Appending a node above the root of a chain produces a longer chain, and appending a node below its deepest leaf also produces a longer chain. In either case, the underlying undirected graph remains a tree, so efficiency remains one.

For **UE*** and **DE***, take two node-disjoint chains of the same size and add a single edge from the leaf of the upper chain to the root of the lower chain. The resulting weakly connected graph is again a tree. The two input indices and the attached index are all one, so the dominance premise holds with equality while the required strict conclusion fails.

5.3.4 Least upper boundedness

Krackhardt's least upper boundedness index evaluates whether pairs of nodes admit a unique lowest common supervisor.

Let h have weakly connected components of sizes $n_1, \dots, n_q$. For nodes i and j in the same component, let

$$B(i, j) = \{b \in V(h) : b \rightsquigarrow i \text{ and } b \rightsquigarrow j\},$$

where reachability is taken to be reflexive for this definition. Let $M(i, j)$ contain the nodes $b \in B(i, j)$ for which no distinct $b' \in B(i, j)$ satisfies $b \rightsquigarrow b'$. Thus, $M(i, j)$ is the set of common supervisors closest to i and j . The pair (i, j) has a unique least upper bound when $M(i, j)$ is a singleton.

Define

$$\text{LUB}(h) = \frac{|\{(i, j) : i \neq j, i, j \text{ lie in the same weak component, and } |M(i, j)| = 1\}|}{\sum_{c=1}^q n_c(n_c - 1)},$$

whenever the denominator is positive.

The LUB index satisfies **A** because relabeling preserves reachability and least-upper-bound structure.

Replication. Replication multiplies both the numerator and denominator by the number of copies. Hence,

$$\text{LUB}(h^k) = \text{LUB}(h)$$

whenever both sides are defined. Therefore, LUB satisfies **IRP** and violates **ERP** and **CRP** on its domain.

Subordination removal and extensions. The LUB index violates **SR**. Let h be the three-node chain

$$j \rightarrow i \rightarrow k.$$

Remove the non-redundant edge (i, k) . The resulting hierarchy h' has edges

$$j \rightarrow i \quad \text{and} \quad j \rightarrow k.$$

In both hierarchies, every ordered pair of distinct nodes in the weak component has a unique least upper bound. Therefore,

$$\text{LUB}(h') = \text{LUB}(h) = 1,$$

contradicting the strict part of **SR**.

The LUB index also violates **UE**, **DE**, **UE***, and **DE***. Appending a chain above or below another chain produces a longer chain, and every relevant ordered pair continues to have a unique least upper bound. Hence, the index remains one and fails the strict conclusions of **UE** and **DE**.

For either conditional extension axiom, take one chain as the base and another node-disjoint chain as the appended hierarchy. Add a single edge from the leaf of the upper chain to the root of the lower chain. The resulting hierarchy is again a chain and has LUB equal to one. The dominance premise in **UE*** and **DE*** holds with equality, but the attached hierarchy does not receive a strictly larger value.

The denominator defining LUB is zero when every weak component is a singleton. The index is therefore undefined on flat hierarchies unless an additional convention is imposed.

5.4 Summary of axiomatic compliance

Table 4 summarizes the compliance of the literature indices with the present axiomatic framework.

Table 4: Axiomatic compliance of hierarchy indices from the literature

Axiom	Avg. Depth	GRC	Connected-ness	Graph Hierarchy	Graph Efficiency	LUB
A	✓	✓	✓	✓ [†]	✓	✓
IRP	✓	×	×	✓ [†]	✓	✓
ERP	×	×	×	×	×	×
CRP	×	✓	✓	×	×	×
SR	✓	×	×	×	×	×
UE* / DE*	✓	×	×	×	×	×
DE	✓	×	×	×	×	×
UE	×	×	×	×	×	×

Notes: ✓ denotes satisfaction and × denotes violation. For indices not defined on all of $\mathcal{H}$, compliance is evaluated whenever all index values required by the relevant axiom are defined. The combined **UE*** / **DE*** row is used because the two axioms have the same status for every index listed. [†]Graph hierarchy is constant on the non-flat DAG domain, so it satisfies **A** and **IRP** trivially. Several measures are undefined on some flat or small hierarchies; see the corresponding subsections.

Sources: Average depth: [Carbonell-Nicolau \(2025b\)](#); GRC: [Mones et al. \(2012\)](#); graph-theoretic dimensions: [Krackhardt \(1994\)](#).

6 Conclusions

This paper distinguishes three interpretations of hierarchy across populations of different sizes. Average indices measure hierarchy per person, total indices measure aggregate hierarchical structure, and concentration indices measure hierarchical structure relative to the number of potential pairs of individuals. These interpretations imply different replication rules. They correspond, respectively, to **IRP**, **ERP**, and **CRP**, subject to positivity in the total and concentration cases. Because these principles impose conflicting rankings under replication, the choice of aggregation rule is part of the measurement objective rather than a neutral technical decision.

The analysis focuses on depth–span indices. A local evaluation function assigns a score according to each individual’s number of supervisors and subordinates, while the population normalization determines how those scores are aggregated. This separates the evaluation of individual vertical position from the treatment of organizational size.

For depth–span average indices, unrestricted upward extension conflicts with subordination removal: no such index satisfies both **UE** and **SR**. This conflict does not extend to every other extension axiom. Weak coordinatewise monotonicity and uniform strictness in at least one coordinate guarantee **SR**, **UE***, and **DE***. If the index also admits a weakly coordinatewise increasing representation satisfying the downward-boundary condition, it satisfies **DE**. The average depth index satisfies these conditions, illustrating the asymmetry between upward and downward extensions.

For concentration indices, the incompatibility with extension axioms is stronger. If a depth–span concentration index takes a positive value on some hierarchy, it satisfies neither **UE*** nor **DE***, and neither **UE** nor **DE**. These impossibilities require no monotonicity, sign, boundedness, or marginal-growth condition on the local evaluation function. They follow from pair-based dilution. For conditional extensions, duplicating a large positive hierarchy creates a dilution effect that a single cross-edge cannot offset. For unrestricted extensions, the axioms impose bounds on unnormalized average scores that drive the pair-normalized values of expanding hierarchies below a fixed positive baseline. Concentration indices can nevertheless satisfy **A**, **SR**, and **CRP** under the stated normalization, monotonicity, and strictness conditions.

Depth–span total indices exhibit the opposite pattern. If the local evaluation has a zero baseline, is weakly increasing in both coordinates, and is uniformly strictly increasing in at least one coordinate, the resulting index satisfies **ERP**, **SR**, **UE**, **DE**, **UE***, and **DE***. Without a population denominator, an appended hierarchy contributes a nonnegative score, and the new cross-edges create a strict gain. Under these conditions, every partial attachment raises the total index above each input hierarchy without using the dominance premise in the conditional extension axioms.

The comparison yields a clear pattern. Average normalization preserves replication invariance but permits population dilution. Under the stated conditions, total aggregation avoids dilution and supports every extension axiom. Pair-based concentration normalization measures hierarchy relative to the number of potential pairs but is incompatible with universal extension monotonicity whenever the index is positive on some hierarchy. No single aggregation rule captures hierarchy per person, aggregate hierarchy, and concentration of hierarchy simultaneously.

The comparison with indices from the literature reinforces these distinctions. Average depth fits the depth–span average family and satisfies **IRP**, **SR**, **UE***, **DE***, and **DE**, but not **UE**. Global reaching centrality and connectedness satisfy **CRP** but violate subordination removal and the extension axioms. Krackhardt’s graph hierarchy index is constant on non-flat directed acyclic graphs, while graph efficiency and least upper boundedness are replication-invariant but insensitive to structural changes for which the present axioms require a strict response. These measures capture graph properties other than the depth–span conception studied here; their classifications clarify the differences among the corresponding measurement objectives.

The main conclusion is that population normalization determines more than scale. It governs replication behavior and constrains the structural monotonicity an index can satisfy. An analyst must therefore specify whether the objective is to measure hierarchy per person, in total, or relative to the number of potential pairs before choosing a functional form. Once that objective is fixed, the results identify the compatible replication, subordination-removal, and extension requirements.

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